

Separating Rate from Composition: A Zero-Loss Decomposition of Electoral Change

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Abstract

Decomposing electoral change — separating shifts in group preferences from shifts in demographic composition — is central to understanding election outcomes. Voters’ group identities structure electoral behavior, yet existing decomposition methods — whether regression-based or the most recent non-linear approaches — introduce systematic error, silently misassigning votes across components and producing residuals exceeding 15% of the total change being explained. I introduce the Rate, Composition, and Volume Decomposition (RCVD), a zero-loss framework that partitions total vote change into three components: changes in group loyalty (rate), group size (composition), and overall participation (volume). Through sequential rather than simultaneous calculation, RCVD eliminates residual error while preserving substantive interpretability. Applied to U.S. presidential elections from 2016 to 2024, RCVD reveals that the 2024 result was driven by a decisive decoupling of rate and composition effects among Hispanic voters. Demographic growth favored Democrats compositionally; rightward shifts in Hispanic vote margins neutralized that advantage entirely.

1 Introduction

After every U.S. presidential election, the post-mortems begin immediately. Political scientists, journalists, campaign strategists, and party leaders debate and champion ex-

planations for the outcome: which groups changed their minds, which grew as a share of the electorate, and what it all means for the future. These explanations are consequential. They shape how parties allocate resources, which voters they court, and how scholars understand democratic change. Recent cycles illustrate this high-stakes pattern: the 2012 Republican “autopsy” emphasized immigration reform to court Latino voters (Republican National Committee, 2013); the 2016 cycle ignited furor over white working-class defections from the Democratic party, linked to both economic disruption (Autor et al., 2020) and identity-based realignment (Sides et al., 2019); and the 2024 election prompted a flood of research into Hispanic voters’ rightward shift (Equis Research, 2024; Catalist, 2025; D. Hopkins and Marble, 2026). Yet existing methods for producing these explanations introduce systematic error — misassigning votes between preference shifts and demographic changes, producing residuals that can exceed 15% of the total change they seek to explain, and obscuring the true drivers of electoral change.

The question at the heart of these post-mortems — whether electoral change reflects shifting preferences or shifting demographics — is foundational to the study of electoral behavior. Voters’ identification with social groups structures their political choices (Campbell, 1960). To the extent people vote according to social group membership, understanding how the composition of these groups shifts over time is key to explaining aggregate election outcomes (Axelrod, 1972). This is particularly relevant as electoral coalitions shift along identity lines (Sides et al., 2019).

The challenge is inherent to the problem’s structure. Current approaches — whether regression-based (Hill et al., 2021) or the recent and widely employed linear difference framework of Marble et al. (2024) (see also Fraga, Velez, et al. 2024) — face a fundamental tension. Separating shifts in group voting preferences from shifts in the demographic composition of the electorate requires either accepting uninterpretable interaction terms or imposing restrictive functional form assumptions. These interactions arise because each group’s vote contribution is the product of its preference margin and its share of the electorate; when both change simultaneously, the combined

effect exceeds the sum of each change measured independently. When scholars omit these interaction terms to maintain interpretability, as is standard practice, the resulting estimates systematically misattribute vote changes. In close elections, residuals of this magnitude can reverse substantive conclusions about which factors drove the outcome. Marble et al. (2024) represents the state of the art, but its decomposition yields a residual of 1.2 million votes — votes that are real but mathematically unassigned to any component.

This paper introduces the Rate, Composition, and Volume Decomposition (RCVD), which resolves this problem through sequential rather than simultaneous calculation. Three terms do all the work: *Rate*, the two-party vote margin within a demographic group; *Composition*, that group’s share of the total electorate; and *Volume*, the total number of voters casting ballots. Each group’s contribution to a candidate’s vote margin is the product of these three quantities. By updating each component sequentially — first allowing rate to shift, then composition, then volume — RCVD accounts for the interactions between components without requiring separate interaction terms. The result is an exact decomposition with zero residual error, and each component retains a clear substantive interpretation.

Applied to U.S. presidential elections from 2016 to 2024, RCVD reveals that the 2024 result was driven by a decisive decoupling of rate and composition effects among Hispanic voters. While demographic growth continued to favor Democrats compositionally, rightward shifts in Hispanic vote margins neutralized that advantage entirely — a pattern invisible to existing methods. Cleanly separating rate from composition is what makes the decoupling visible.

The article proceeds as follows. Section 2 reviews existing decomposition methods and develops the RCVD framework, demonstrating why standard approaches generate residual error. Section 3 presents numerical examples with both hypothetical and real data, validating RCVD through controlled scenarios. Section 4 applies RCVD to U.S. presidential elections from 2016 to 2024, revealing how the method exposes data quality issues and captures Hispanic voting dynamics that existing approaches miss. Section

5 concludes.

2 Historical Approaches and the RCVD Framework

Measuring how electoral coalitions shift across elections requires distinguishing two fundamentally different dynamics: changes in how groups vote, and changes in who comprises the electorate. Political scientists have developed a range of approaches to this problem, each making different tradeoffs between interpretability and precision.

Contemporary scholarship has emphasized the role of identity-based coalitions in shaping electoral outcomes. Achen and Bartels (2016) argues that political science needs to move away from an exclusive focus on individuals and toward greater attention to identity groups. Sides et al. (2019) demonstrates that the 2016 Trump campaign succeeded in part through reshaping party coalitions around identity politics. In this context, Lemi (2021) demonstrates that building electoral coalitions based on group membership has become a central campaign strategy — which means that accurately separating rate from composition effects is not merely a technical exercise but a substantive one.

Recent scholarship has developed various approaches to decompose vote changes into their constituent parts. Hill et al. (2021) uses a regression-based approach to attribute vote share changes to either composition or conversion, while Zingher (2019) measures how many votes each social group contributes to party coalitions across elections, distinguishing demographic from behavioral forces driving those changes. However, regression-based approaches impose a linear specification on what is fundamentally a multiplicative relationship: each group’s vote contribution is the product of its preference margin, its share of the electorate, and overall turnout. A linear model cannot recover the separate factors of a product, and misspecification of this form introduces systematic bias rather than mere imprecision (Wooldridge, 2010). As Engelhardt (2019) points out, shifts in the demographic composition of the electorate and shifts in vote choice within groups can mask each other, making regression coefficients difficult to interpret and obscuring the true drivers of electoral change.

Other analyses have employed simpler extrapolation methods. Fraga, McElwee, et al. (2021) examines how racial differences in turnout affected the 2016 election by applying 2012 turnout rates to construct counterfactual vote shares, isolating the contribution of differential turnout from shifts in vote choice. Carmines et al. (2016) decomposes each group’s contribution to the party coalitions into size, turnout, and loyalty components, using this framework to show how populist voters’ distinctive profile across these dimensions helped deliver the 2016 outcome to Trump. These studies illuminate specific aspects of how rate and composition shifts drive electoral change, but neither provides a framework that accounts for how those components interact when they move simultaneously. These interactions are crucial — they are the additional effect that arises when two or more components change at the same time: if a group both grows as a share of the electorate and shifts its voting preferences, the combined impact exceeds the sum of each change measured in isolation.

The importance of accurate decomposition extends beyond single elections to forecasting. Calvo et al. (2024) demonstrates that even perfect knowledge of demographic shifts provides insufficient information to guarantee accurate forecasts — cleanly separating shifts in group size from shifts in how groups vote is essential to understanding why forecasts succeed or fail. Grimmer, Knox, et al. (2024) argues that properly evaluating forecasts requires understanding how shifts in group composition and vote choice interact across multiple election cycles. Together these contributions establish that the interaction between rate and composition effects is not a methodological curiosity but a substantive problem with direct consequences for how well we understand and anticipate electoral outcomes.

The most comprehensive framework to date is Marble et al. (2024), which rejects the linearity assumption of earlier regression-based approaches and adopt a non-linear specification that directly models how each group’s vote margin (rate) and electoral share (composition) changes between elections. Explicitly, rate is a net measure: if 60% of a group votes Republican and 40% votes Democrat, the margin rate is +20% (or +0.20), meaning that group contributes a net positive margin to Republicans. Marble

et al. (2024) captures each group’s contribution to total vote change as:

$$\begin{aligned}
\text{Diff Net}_{t,t-1}(x) = & [\text{Vote Share}_t(x, \text{Republican}) - \text{Vote Share}_t(x, \text{Democrat})] \\
& \times \text{Turnout}_t(x) \times \text{Group Size}_t(x) \\
& - [\text{Vote Share}_{t-1}(x, \text{Republican}) - \text{Vote Share}_{t-1}(x, \text{Democrat})] \\
& \times \text{Turnout}_{t-1}(x) \times \text{Group Size}_{t-1}(x)
\end{aligned} \tag{1}$$

Where $\text{Turnout}_t(x) \times \text{Group Size}_t(x)$ represents the group’s share of total votes cast. Here x indexes the group of interest and t indexes elections rather than years.¹ To separate the rate effect from the composition effect, they calculate each component independently: first fixing group composition (in their terminology, the combined turnout and group size term) at the baseline election and measuring the impact of changed vote shares, then fixing vote shares at the baseline and measuring the impact of changed composition. The two components are then summed to estimate each group’s total contribution to electoral change. Fraga, Velez, et al. (2024) applies this same decomposition framework to Latino subgroups, finding that Trump’s 2020 gains reflected shifts in vote choice concentrated among subgroups already predisposed toward Republicans — rather than a uniform shift across Latino voters broadly.

However, what I term the Linear Difference Decomposition (LDD) faces two key challenges. First, it conflates composition and volume — treating them as a single combined term rather than separating them. In RCVD, composition refers strictly to the relative share of the electorate each group comprises (for example, 60 percent White, 40 percent non-White), while volume is the total number of voters casting ballots. The LDD combines these into a single term — group size multiplied by turnout rate — which captures whether a group contributed more or fewer votes overall, but cannot distinguish whether that change came from the group becoming a larger share of the electorate or from the electorate simply expanding. Consider turnout in the 2008 presidential election: turnout reached a 40-year high (Woolley and Peters, 2021),

¹i.e. the 2016 U.S. presidential election might be $t = 0$ and 2020 might be $t = 1$.

but the aggregate increase masked important differential shifts. Turnout among Black voters rose by nearly 5 percent while turnout among White voters declined by just over 1 percent (Lopez and Taylor, 2009). The LDD would capture positive effects for Black voters and negative effects for White voters, but by conflating group share with total electorate size, it understates the compositional shift. Black voters became a meaningfully larger share of the electorate, not just a group that turned out more.

Working in percentages alone eliminates the need for a separate volume term — if researchers care only about relative contributions rather than absolute vote counts, normalization is sufficient, and the first challenge is resolved. However, the second challenge remains regardless of normalization: even in percentage terms, calculating rate and composition effects independently and summing them produces a residual whenever both components change simultaneously. The standard approach is not merely incomplete in the quantities it reports; it misattributes the quantities it does capture.

The second — and more fundamental — challenge is that calculating rate and composition effects independently and then summing them fails to account for their interaction. When both rate and composition change simultaneously between elections, the total effect is not simply the sum of the two independent effects. To see why, consider a single group: its vote outcome is the product of its margin rate $r(t)$ and its electoral share $c(t)$, giving $z(t) = r(t)c(t)$. Because this relationship is multiplicative rather than additive, changes in rate and composition interact: fixing one component at its baseline value while varying the other yields a valid partial effect, but the sum of these partial effects will not equal the total change whenever both components move simultaneously. The LDD calculates the rate effect by fixing composition at t_1 and allowing rate to shift: $[r(t_2) - r(t_1)]c(t_1)$. It then calculates the composition effect by fixing rate at t_1 and allowing composition to shift: $r(t_1)[c(t_2) - c(t_1)]$. These two terms are then summed to estimate total change — but as the next equation shows, this sum is missing something.

The actual total change is $\Delta z(t) = r(t_2)c(t_2) - r(t_1)c(t_1)$. Expanding this alge-

braically reveals the missing term:

$$\begin{aligned}
\Delta z(t) &= z(t_2) - z(t_1) \\
&= r(t_2)c(t_2) - r(t_1)c(t_1) \\
&= [r(t_2) - r(t_1)]c(t_1) + r(t_1)[c(t_2) - c(t_1)] + [r(t_2) - r(t_1)][c(t_2) - c(t_1)] \quad (2)
\end{aligned}$$

The first two terms match the LDD calculation, but the third term— $[r(t_2) - r(t_1)][c(t_2) - c(t_1)]$ —represents an interaction effect. This interaction term captures the additional votes generated (or lost) when both rate and composition change together. Substantively, this interaction represents a critical dynamic: when a group both grows as a share of the electorate and shifts its rate, the impact is multiplicative rather than simply additive. For instance, if Hispanic voters both increase as a share of the electorate and shift toward Republicans, the net effect is larger than simply adding the two effects independently would suggest.

To illustrate concretely with the example developed in Section 3: Group A shifts from a +20% margin rate to −10% while growing from 30% to 33.3% of the electorate. The standard approach measures the composition effect using the *old* rate (+20%), asking what those extra voters would have contributed if they still favored Candidate X. But Group A now opposes Candidate X. Measuring composition at the old rate gets the direction wrong — it credits Candidate X with votes that actually went to Candidate Y. The interaction between the rate reversal and the compositional growth is exactly what falls into the residual.

When the LDD omits this interaction term, the residual must go somewhere. Omitting it implicitly assumes the interaction is zero or trivially small — but as Section 4 demonstrates, it can represent millions of votes in real elections. In the case of Marble et al. (2024)’s analysis of 2012 to 2016, this produces a 1.2-million-vote residual — roughly 15% of the total change being decomposed. The problem compounds when volume enters the picture. With three components — rate, composition, and vol-

ume — the complete decomposition requires not just three main effects but also three two-way interactions and one three-way interaction:

$$\begin{aligned}
\Delta z(t) &= z(t_2) - z(t_1) \\
&= r(t_2)c(t_2)v(t_2) - r(t_1)c(t_1)v(t_1) \\
&= [r(t_2) - r(t_1)]c(t_1)v(t_1) + r(t_1)[c(t_2) - c(t_1)]v(t_1) + r(t_1)c(t_1)[v(t_2) - v(t_1)] \\
&\quad + [r(t_2) - r(t_1)][c(t_2) - c(t_1)]v(t_1) + [r(t_2) - r(t_1)]c(t_1)[v(t_2) - v(t_1)] + r(t_1)[c(t_2) - c(t_1)][v(t_2) - v(t_1)] \\
&\quad + [r(t_2) - r(t_1)][c(t_2) - c(t_1)][v(t_2) - v(t_1)]
\end{aligned} \tag{3}$$

The equation’s first line contains the three main effects — what the LDD attempts to calculate. The second and third lines contain the interaction terms: the products of simultaneous changes across components. Prior work has not directly calculated volume as a separate component, implicitly treating any unaccounted votes as a Volume effect rather than isolating it explicitly. However, the underlying issues exist whether the analysis is percentage-based or considers total votes.

As a result, methods that rely only on calculating the individual component effects do not account for how these components interact when they change simultaneously. This generates error because the LDD specification is, in essence, treating the problem as if changes in each component can be evaluated independently, when in fact the underlying relationship is multiplicative. If rate and composition changes are small, or if they move in offsetting directions, the interaction terms may be relatively minor and the LDD provides a reasonable approximation. However, as the magnitude of changes grows-or when rate and composition shifts reinforce each other-the interaction terms become substantial, and the LDD approximation deteriorates. The challenge is illustrated in Figure 1, which shows how approximations that ignore interaction effects accumulate error as the changes being measured become larger. In the electoral setting, this means decomposition error grows precisely when demographic and preference shifts are largest — the cases of greatest substantive interest.

RCVD resolves the interaction term problem through sequential rather than simultaneous calculation. The underlying intuition is straightforward: when both rate and

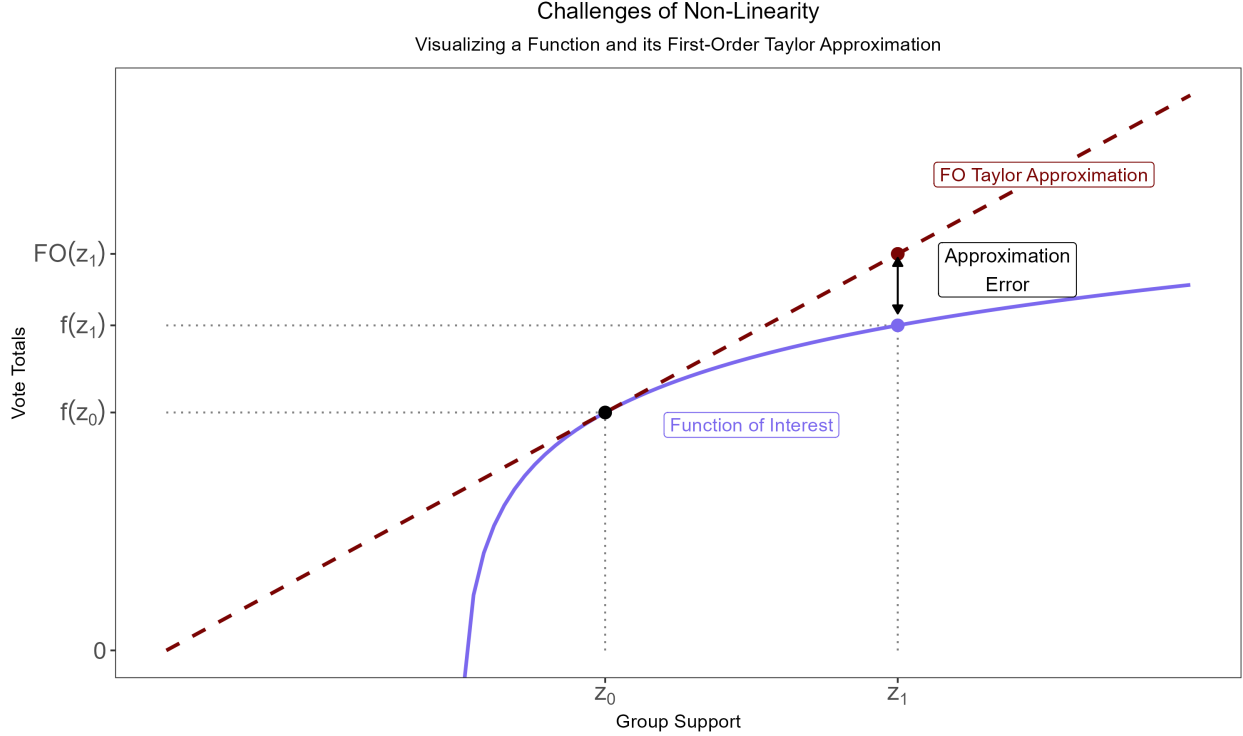


Figure 1: This figure illustrates the core problem with additive decompositions. The curved line represents a group’s actual vote contribution — the product of its rate and composition — while the straight line represents the additive approximation used by the LDD. When changes are small (near z_0), the approximation is close. But as rate and composition shift further (toward z_1), the gap between the true multiplicative relationship and the additive approximation grows. In electoral terms, this means decomposition error is smallest when little changes between elections and largest when demographic and preference shifts are substantial — precisely the cases of greatest substantive interest. Based on Figure 4.3.1 in Reed (1998).

composition change between elections, we cannot meaningfully measure the impact of composition changes while holding rate fixed at its prior values. As Citrin et al. (2003) demonstrates, the composition of the voter pool — who actually turns out — can shift substantially across elections. Measuring composition effects using outdated rates — that is, asking what the new voter pool would have produced if the old voting patterns had held — introduces bias that is not merely a rounding error: it can reverse the estimated sign of a composition effect when a candidate crosses from winning to losing a group between elections.

RCVD avoids this problem by updating each component before measuring the next. First, it asks: holding the electorate’s size and shape fixed, what is the impact of changed voting preferences? Then, taking those new preferences as given, it asks: what

is the impact of the electorate’s changed shape? Because the composition calculation already uses the updated rate, the interaction between the two is absorbed naturally — there is no residual to assign. The sequential approach asks the right counterfactual question at each step: not “what would have happened if only one thing changed,” but “given that preferences have already shifted, what is the additional impact of compositional change?”

Define $z_i(t)$ as the net margin votes that group i contributes to a candidate, where t indexes elections and i indexes demographic groups. Following Marble et al. (2024), the analysis works with vote margins rather than raw vote totals. Each group’s contribution is the product of three quantities:

$$z_i(t) = r_i(t)c_i(t)v(t) \tag{4}$$

Where $r_i(t)$ is the margin rate at which group i votes for the candidate, $c_i(t)$ is group i ’s share of the total electorate, and $v(t)$ is the total number of voters. Note that $r_i(t)$ and $c_i(t)$ vary across groups while $v(t)$ is constant — the total electorate size is the same for all groups. Note that these shares are constrained: $\sum_i c_i(t) = 1$ for any election t , ensuring that the group-level contributions exhaust the total electorate.

This specification differs from Marble et al. (2024) and Fraga, Velez, et al. (2024) in its explicit treatment of volume as a separate component. Because volume is calculated last in the sequential decomposition, it functions as a pure scaling factor: it captures only the proportional expansion or contraction of the electorate after rate and composition have already been updated to their new values. Isolating volume explicitly matters for two reasons. First, total votes provides the only firm external benchmark for validating survey-based estimates — since group compositions and vote choice are estimated from surveys, the final reported vote tally is the only hard number available for checking plausibility, a point Section 4 demonstrates directly. Second, as Marble et al. (2024) notes, this specification allows researchers to apply exit poll results to vote

tallies almost immediately after an election, before detailed demographic breakdowns are available.

I propose the following sequential decomposition of $\Delta z_i(t)$ into its rate, composition, and volume components:

$$\begin{aligned}\Delta z_i(t) = & [r_i(t_2) - r_i(t_1)]c_i(t_1)v(t_1) \\ & + r_i(t_2)[c_i(t_2) - c_i(t_1)]v(t_1) \\ & + r_i(t_2)c_i(t_2)[v(t_2) - v(t_1)]\end{aligned}\tag{5}$$

The first term isolates the rate effect for group i : the change in $z_i(t)$ when rate shifts from $r_i(t_1)$ to $r_i(t_2)$, holding composition and volume at their initial values. The second term isolates the composition effect: the change in $z_i(t)$ when group i 's electoral share shifts from $c_i(t_1)$ to $c_i(t_2)$, using the updated rate $r_i(t_2)$ but holding volume fixed. The third term isolates the volume effect: the change in $z_i(t)$ when overall turnout shifts from $v(t_1)$ to $v(t_2)$, using both the updated rate and updated composition for group i . The proof that these three terms sum exactly to $z_i(t_2) - z_i(t_1)$ — with zero residual — is in the Appendix. This is the RCVD specification.²

Two features of this specification warrant attention. First, turnout affects both $c_i(t)$ and $v(t)$, while population growth affects $v(t)$ directly and $c_i(t)$ only when it is differential across groups. This raises the question of whether these should be decomposed further. They need not be, for two reasons. Across election cycles just four years apart, underlying changes in the racial composition of the potential electorate are unlikely to be substantively meaningful compared to turnout shifts.³ More fundamentally, turnout changes only matter insofar as they are differential across groups.

If all groups increase turnout at the same rate, relative group sizes are preserved and

²RCVD operates on group-level aggregates, not individual voters. If a voter who shifted preference does not turn out in the subsequent election, the rate component reflects the preferences of those who did vote, while the volume component captures the change in total participation.

³On average since 2010, the Hispanic population in the U.S. has only increased as a percentage of the total population by roughly 1% every four years (U.S. Census Bureau, 2024).

there is no compositional effect. Universal changes — overall population growth, uniform turnout increases — are absorbed into volume. Only differential changes shift composition.

A second crucial feature is that this sequence is not unique, and the choice of ordering carries substantive implications. The decomposition can be performed in any order of rate, composition, and volume. Different orderings yield different component estimates because each ordering implicitly defines a different counterfactual comparison. This is a deeper point than it might appear — every decomposition approach embeds assumptions about ordering. The LDD simply hides this by implicitly treating all cross-partial terms as trivially small. RCVD makes that assumption explicit and then eliminates the need for it entirely.

The rate $\rightarrow$ composition $\rightarrow$ volume ordering is the most interpretable for two reasons. First, calculating rate effects on the first period’s composition and volume ensures continuity with existing literature — the rate calculation matches prior work exactly, correcting its downstream errors without redefining them. Second, calculating volume last means it functions as a pure scaling factor: it captures only the proportional expansion or contraction of the electorate after rate and composition have already been updated to their new values. To illustrate concretely: the rate calculation asks how the 2016 electorate would have voted if only preferences changed — the same question prior work asks. The composition calculation then asks, given those new preferences, what happens when the electorate shifts from 71% White to 68% White? And the volume calculation asks, given both the new preferences and the new demographic mix, what happens when 26 million additional voters show up? Section 3 demonstrates how alternative orderings change the interpretation of each component and develops the temporal logic underlying the chosen sequence.

3 Proof of Concept: Validation through Controlled Simulation

This section moves from theoretical framework to empirical application. The previous section introduced the mathematical structure of RCVD and explained why sequential calculation resolves the interaction term problem that affects the Linear Difference Decomposition. Here, I demonstrate how these approaches perform in practice. I begin with a simple numerical example using hypothetical data to illustrate exactly how the LDD generates a residual while RCVD achieves zero error. I then apply both methods to increasingly complex scenarios: first, a two-group analysis of the 2016-2020 election showing the magnitude of the LDD’s residual in real data; second, a controlled scenario isolating the rate-composition interaction by holding volume constant; and finally, a justification for the rate $\rightarrow$ composition $\rightarrow$ volume calculation sequence based on theoretical interpretability.

A Simple Numerical Example

To see precisely how the LDD generates a residual while RCVD does not, consider a straightforward hypothetical scenario with two voter groups across two elections. Following Marble et al. (2024), I work with vote margins—the difference between votes for Candidate X and votes for Candidate Y within each group. Table 1 presents the data, using round numbers throughout to allow readers to verify each calculation step.

Table 1: Hypothetical Two-Group Election Data

Group	Votes for X	Votes for Y	Total	X Margin	X Margin Rate	% of Total
<i>Panel A: Election 1</i>						
Group A	18	12	30	+6	+20%	30.0%
Group B	28	42	70	−14	−20%	70.0%
Total	46	54	100	−8	−8%	100%
<i>Panel B: Election 2</i>						
Group A	18	22	40	−4	−10%	33.3%
Group B	36	44	80	−8	−10%	66.7%
Total	54	66	120	−12	−10%	100%

Margin = Votes for X − Votes for Y. Margin Rate = Margin $\div$ Total Voters.

In Election 1, Group A favors Candidate X (18 votes for X versus 12 for Y, yielding a +6 vote margin and a +20% margin rate), while Group B favors Candidate Y (28 votes for X versus 42 for Y, yielding a -14 vote margin and a -20% margin rate). The total margin across both groups is -8 votes, meaning Candidate Y leads by 8 votes overall. By Election 2, both groups shift toward Candidate Y. Group A now opposes X (18 votes for X versus 22 for Y, yielding a -4 vote margin and a -10% margin rate), and Group B continues opposing X but less strongly (36 votes for X versus 44 for Y, yielding an -8 vote margin and a -10% margin rate). The total margin is now -12 votes, meaning Candidate X's margin declined by 4 votes between the two elections. This change resulted from three factors: shifts in margin rates within groups (rate), changes in group sizes (composition), and an increase in overall turnout (volume).

Table 2 shows how the LDD and RCVD decompose this 4-vote margin change.

Table 2: LDD vs RCVD Decomposition of 4-Vote Margin Change

Component	LDD Calculation	RCVD Calculation
<i>Rate Effect:</i>		
Group A	$(-0.30) \times 0.30 \times 100 = -9.0$	$(-0.30) \times 0.30 \times 100 = -9.0$
Group B	$(+0.10) \times 0.70 \times 100 = +7.0$	$(+0.10) \times 0.70 \times 100 = +7.0$
Total	-2.0 votes	-2.0 votes
<i>Composition Effect:</i>		
Group A	$(+0.20) \times (+0.033) \times 100 = +0.667$	$(-0.10) \times (+0.033) \times 100 = -0.333$
Group B	$(-0.20) \times (-0.033) \times 100 = +0.667$	$(-0.10) \times (-0.033) \times 100 = +0.333$
Total	+1.333 votes	0.0 votes
<i>Volume Effect:</i>		
Group A	$(+0.20) \times 0.30 \times 20 = +1.2$	$(-0.10) \times 0.333 \times 20 = -0.667$
Group B	$(-0.20) \times 0.70 \times 20 = -2.8$	$(-0.10) \times 0.667 \times 20 = -1.333$
Total	-1.6 votes	-2.0 votes
Sum of Components	-2.267 votes	-4.0 votes
Actual Change	-4.0 votes	-4.0 votes
Residual Error	-1.733 votes	0.0 votes

RCVD running totals: -2 votes after Rate, -2 after Composition, -4 after Volume

Both methods calculate the same rate effect (-2.0 votes), but they diverge in the composition and volume calculations. The LDD holds all components at Election 1 values when calculating each effect independently. For the composition effect, it applies Election 1's margin rates (+20% for Group A, -20% for Group B) to the compositional

shift (+3.3 percentage points for Group A, -3.3 for Group B), producing a +1.333 vote effect. For the volume effect, it applies Election 1’s margin rates and composition to the 20-voter increase, producing a -1.6 vote effect. When summed, these components total -2.267 votes, leaving a -1.733 vote residual.

In contrast, RCVD calculates each component sequentially, using updated values at each step. After the rate effect reduces the margin from -8 to -10 votes, the composition calculation uses the *new* margin rates (-10% for both groups) rather than the old rates. Because both groups have identical margin rates in Election 2, the compositional shift from 30%/70% to 33.3%/66.7% produces zero net effect—moving voters between two groups that vote identically has no impact on the overall margin. The volume effect then scales from 100 to 120 voters using both the new margin rates and new composition, reducing the margin by an additional 2.0 votes (from -10 to -12). The RCVD components sum exactly to -4.0 votes with zero residual.

This simple example demonstrates why RCVD is necessary: even with straightforward data and small changes, the LDD generates residuals because it cannot properly account for how changes in rate, composition, and volume interact. The residual arises specifically from the LDD measuring composition at the old rate (+20%) when the actual rate has shifted to -10% — the same mismatch identified in Section 2. In real elections with larger shifts and more groups, these residuals grow substantially. The Two-Group Simulation in the Appendix confirms this pattern with real election data, showing that the LDD produces a residual exceeding 1.4 million votes — roughly 15% of the total change — even in a simplified two-group analysis of the 2016-2020 election.

Non-Uniqueness and the Fixed RCVD Path

As noted in the conclusion of Section 2, the Rate, Composition, and Volume Decomposition (RCVD) can be calculated in any order. Mathematically, there are six possible permutations of the rate (ΔR), composition (ΔC), and volume (ΔV) terms that satisfy the zero-loss property of the RCVD framework. This path-dependence is inherent to correctly modeling the interaction effects that the LDD omits. However, different

orderings yield different interpretations of each component’s contribution, as each ordering implicitly defines a different counterfactual comparison. Therefore, the chosen order ($\Delta R \rightarrow \Delta C \rightarrow \Delta V$) requires substantive justification.

Figure A4 illustrates an alternative zero-loss order (composition $\rightarrow$ rate $\rightarrow$ volume) when applied to the simple 2016-2020 simulation. While the total MAE remains 0, the calculated contribution of rate and composition shifts: the alternative order assigns a slightly larger role to composition and a slightly smaller role to rate.

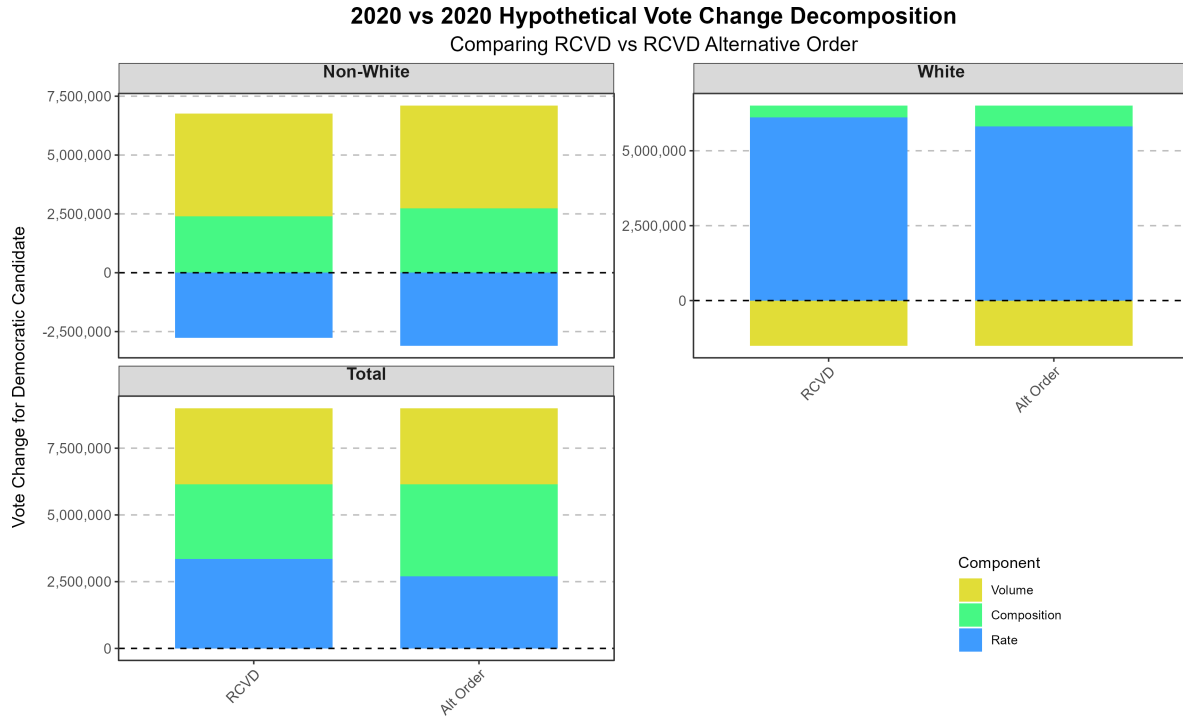


Figure 2: This figure shows that shifting the order in which RCVD is calculated yields no error term, but can shift the interpretation of the impacts of both rate and composition.

The $\Delta R \rightarrow \Delta C \rightarrow \Delta V$ ordering provides the most interpretable decomposition for two reasons. First, this order ensures continuity with existing literature. The convention in electoral studies has always been to calculate the change in group loyalty (rate) based on the prior period’s electorate composition. By sequencing rate first, the ΔR term captures preference changes holding composition and volume at their initial values. This means the rate calculation in RCVD produces exactly the same value as in the LDD, ensuring that the large body of existing work focused on rate changes remains valid while the composition and volume terms are corrected.

Second, this order ensures that volume functions as a proportional scaling factor. By sequencing volume last, we calculate it after rate and composition have already been updated. This means volume captures only the impact of expanding or contracting the electorate while preserving the relationships established by the new rate and new composition. Intuitively, if we shifted composition first (say, increasing the non-White share of the electorate) before calculating volume, the volume effect would be confounded with compositional changes. By placing volume last, it represents a clean proportional increase or decrease in the system’s size, distributed across the electoral configuration determined by rate and composition.

Beyond these methodological considerations, the ordering reflects the temporal structure of electoral change. Preference shifts — captured by rate — evolve over years through lived experience, media environments, and gradual attitude change. The composition of the active electorate is shaped closer to the election through registration drives, mobilization efforts, and differential turnout decisions, often made in response to existing group preferences. Volume — total participation — is resolved last, determined essentially on Election Day by enthusiasm, convenience, and local conditions. Sequencing the decomposition in this order thus mirrors the causal sequence through which electoral outcomes are produced. In sum, the $\Delta R \rightarrow \Delta C \rightarrow \Delta V$ ordering ensures that each component has a clear, theoretically meaningful interpretation that aligns with how electoral scholars conceptualize these dynamics.

4 Real-World Application: ANES, Pew, and the 2024 Hispanic Vote

The preceding section established the RCVD framework and demonstrated that the Linear Difference Decomposition obscures the mechanisms driving electoral change. This section applies RCVD to recent U.S. presidential elections, with particular attention to the Hispanic electorate — the largest and fastest-growing non-White demographic group, and the subject of intense recent debate about whether its electoral impact reflects demographic growth or shifting preferences (Fraga, Velez, et al., 2024;

Corral and Leal, 2024).

The analysis proceeds in three parts, each serving a distinct purpose. First, I apply RCVD to the 2020 election using ANES data (ANES, 2021), showing how it reveals different interpretations of compositional effects compared to LDD methods. Second, I compare ANES and Pew data for the 2020 election, demonstrating that Pew provides greater descriptive accuracy for analyzing Hispanic voter behavior. This comparison supports recent findings by Fraga, Velez, et al. (2024) on Hispanic electoral volatility. Finally, I analyze three electoral cycles (2016–2020, 2020–2024, and 2016–2024) using Pew data, isolating how shifts in voter preferences (rate), demographic composition (composition), and overall turnout (volume) shaped the 2024 election outcome. Analyzing multiple cycles is necessary because RCVD decomposes change between pairs of elections; the three-cycle structure reveals whether the 2024 outcome reflects a single-cycle shock or a cumulative trend.

The 2016 to 2020 Baseline: Compositional Effects in the ANES

Political analysis of the 2020 presidential election typically relies on changes in two-party vote margins within demographic groups.⁴ However, a change in a group’s aggregate vote margin may result from a shift in partisan preference (rate), a change in that group’s share of the electorate (composition), or overall turnout changes (volume). Standard decomposition methods fail to separate these effects cleanly, hindering our understanding of what drove the election outcome.

I begin by applying the RCVD framework to data from the 2020 American National Election Studies (ANES) (ANES, 2021), which represents the discipline’s historical “gold standard.” While ANES has limitations in describing recent election outcomes (discussed in the next subsection), it provides a useful starting point for demonstrating RCVD’s value.

Conventional analysis suggests that the Democratic victory was driven by a modest

⁴Following Marble et al. (2024), I work with two-party vote margins throughout this analysis. A margin represents the difference in vote shares between parties: for example, if Democrats receive 72% and Republicans 28% of a group’s votes, the Democratic margin is +44 percentage points. While I report some results in terms of vote shares for interpretability, the RCVD framework decomposes margin changes.

rate shift among non-White voters combined with significant compositional advantages ([pew_behind_2021](#); Bump, 2021). The RCVD decomposition reveals a more nuanced picture. By correctly partitioning the total vote shift, I find that compositional effects are significantly less beneficial to Democrats than LDD analyses suggest. Figure 3 shows how the LDD overstates the positive impact of compositional shifts for Democrats by roughly 365,000 votes.

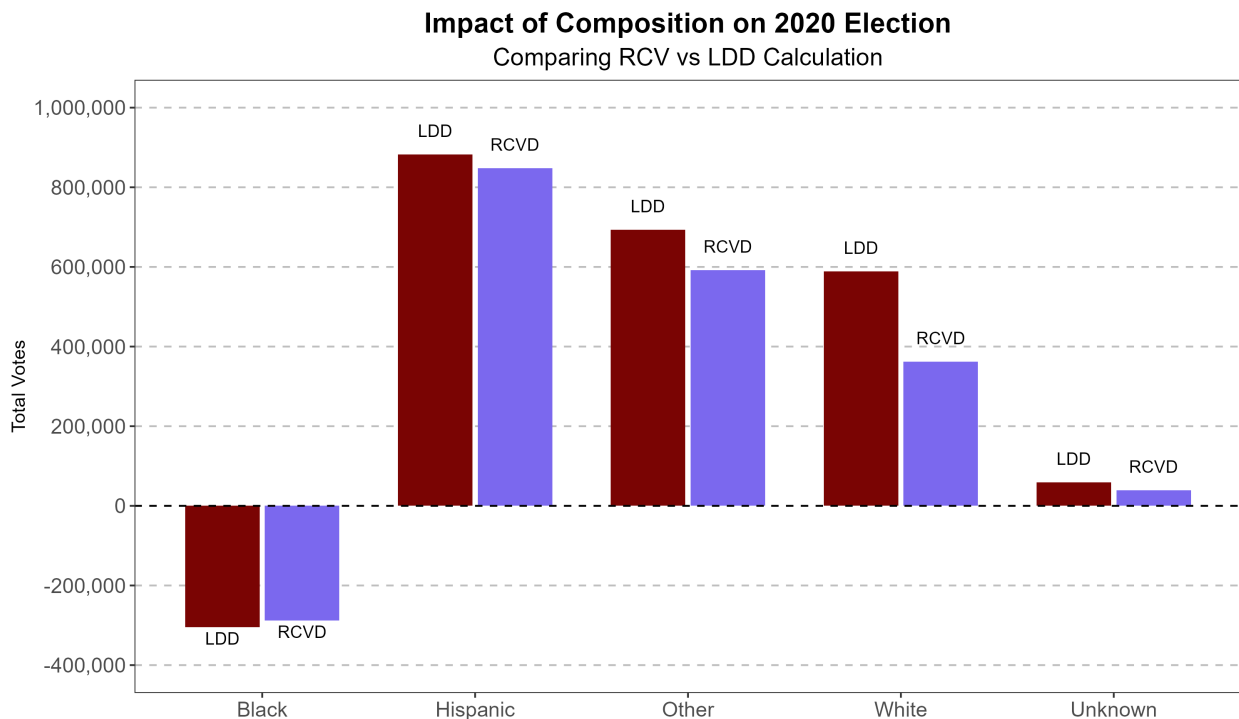


Figure 3: This figure shows how shifting from the LDD calculation to the RCVD calculation changes the evaluation of the impact of composition on the 2020 election. In particular, the LDD calculation overemphasizes the benefits of composition towards Democrats by several hundred thousand votes.

The RCVD framework provides a zero-loss allocation, ensuring that the sum of the rate, composition, and volume components precisely equals the observed overall change in the national vote margin. This contrasts with LDD methods that leave unexplained residual terms. Figure 4 shows that for White voters, the LDD’s error actually exceeds the total impact of composition on the final outcome. The total error across all groups is roughly 700,000 votes versus a total compositional impact (from RCVD) of roughly 1.5 million votes.

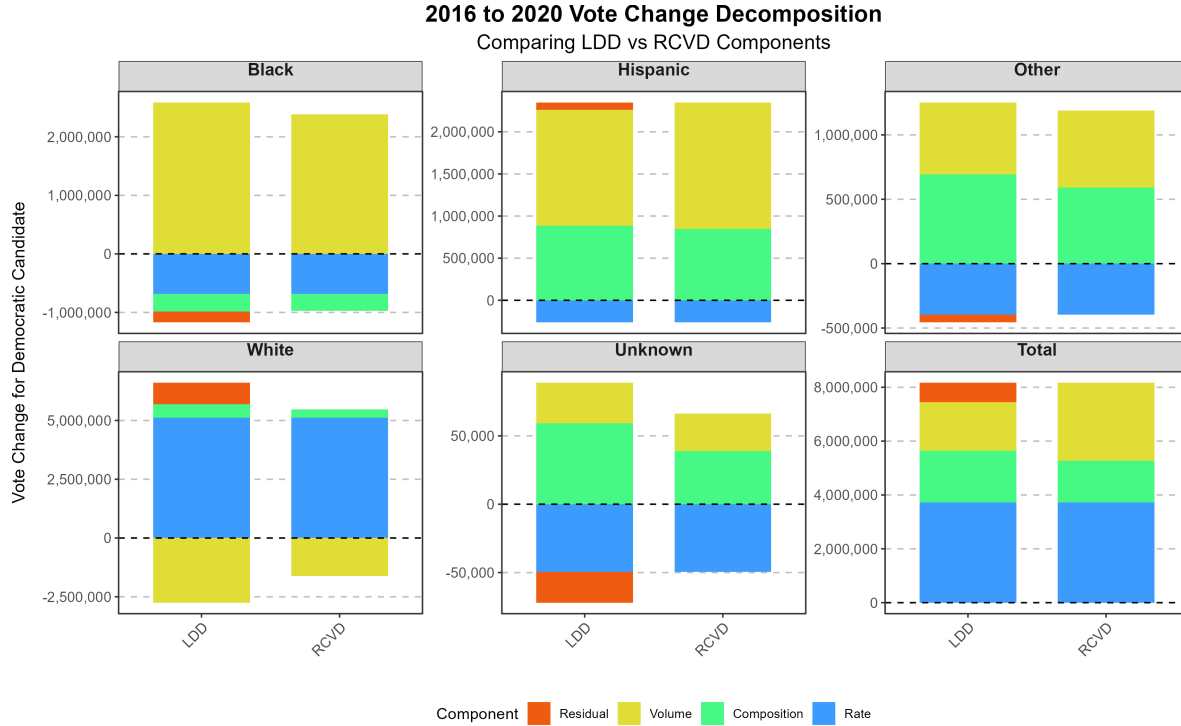


Figure 4: This figure shows how shifting to RCVD shifts interpretations of the 2020 election results when compared to 2016. It highlights that the magnitude of the error associated with the LDD is oftentimes several hundred thousand votes, and in the case of White voters, actually exceeds the total impact of composition.

The application of RCVD to the 2020 ANES data demonstrates that standard methods risk substantial bias in understanding how composition affects election outcomes. This finding motivates the subsequent analysis, where I show that Pew data provides greater accuracy in describing recent elections and use it to analyze the increasingly salient Hispanic electorate.

Survey Divergence: Comparing ANES and Pew

The robustness of any electoral decomposition depends on the descriptive accuracy of the underlying survey data. While ANES remains the discipline’s preferred benchmark, its ability to capture rapidly evolving preferences among the Hispanic electorate has been questioned (Fraga, 2016). I therefore compare ANES to Pew Research Center’s validated voter data (Hartig et al., 2023), which employs a different sampling strategy and may achieve better resolution for non-White subpopulations.

Figure 5 compares the two datasets’ accuracy and their RCVD decompositions for

Hispanic voters. The lower panel shows that ANES overstates Democratic support by nearly 4 million votes, while Pew understates it by under 600,000 votes (Wooley and Peters, 2024)—suggesting Pew provides a more accurate picture of the 2020 electorate. The upper panel reveals a fundamental difference in interpretation. ANES shows a modest Republican gain in rate among Hispanic voters but attributes the overall Democratic increase from this group to composition—a potentially persistent, demographic-driven effect. In contrast, Pew isolates a significantly larger rate effect driving the net impact of Hispanic voters, with composition playing a minimal role. This suggests ANES may underestimate preference volatility among Hispanic voters, supporting the argument in Fraga, Velez, et al. (2024) about the magnitude of Hispanic electoral shifts.

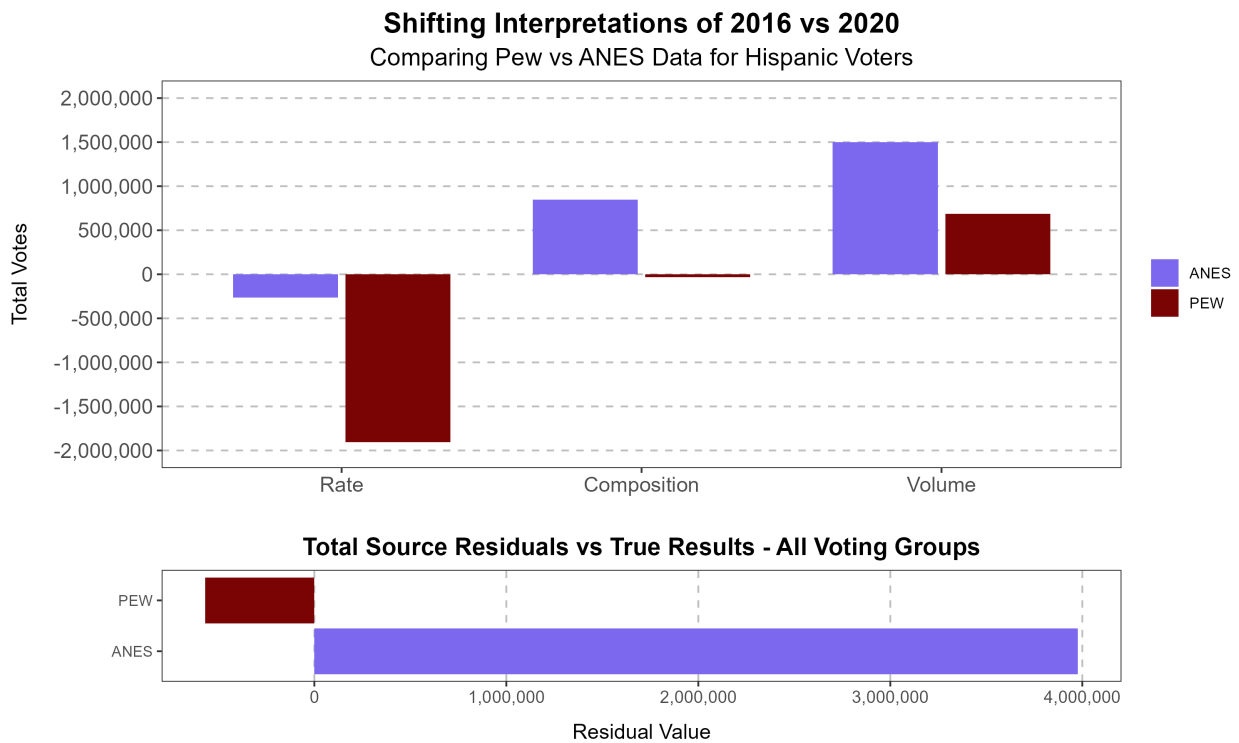


Figure 5: This figure compares the implied election results from ANES data versus Pew data. The upper portion of the figure shows that utilizing Pew data fundamentally alters the interpretation of the impact of Hispanic voters on the 2020 election. In particular, it shows that there is no evidence that Democrats benefited from a shift into Hispanic voters in 2020 and instead suffered a significant rate decline. The lower portion of the figure shows that the implied error in votes of using ANES data to evaluate the 2020 election is nearly 4 million votes, whereas using the Pew data has an error of less than 600,000.

This difference in interpretation carries strategic implications. If the change is mostly compositional (as ANES suggests), the Democratic challenge is primarily organizational. If the change is driven by rate effects (as Pew suggests), the challenge is fundamentally one of persuasion. The Pew data thus refutes the notion that Democrats can rely on demographic changes alone (Klein, 2024; Corral and Leal, 2024), revealing deeper vulnerability to Republican outreach than previously recognized.

This comparison serves as a methodological caution: researchers should carefully assess whether their data reflects reality, as argued in Grimmer, Marble, et al. (2022). By embracing the superior descriptive validity of Pew for the Hispanic electorate, the RCVD framework establishes that Hispanic volatility is driven less by passive demographic turnover and more by active shifts in partisan loyalty—a crucial baseline for understanding the dramatic shifts in the 2024 election.

The Full Picture: Decomposition Across Three Electoral Cycles

I now apply the RCVD framework to analyze vote change across three electoral cycles: 2016–2020, 2020–2024, and the full 2016–2024 interval. This analysis demonstrates how the compounding effects of rate and composition determined the 2024 electoral outcome.

The 2016 to 2020 Cycle— Figure 6

The 2016 to 2020 period represents the Democratic bounce-back from their 2016 loss. While accounting for the translation of the popular vote to the Electoral College involves challenges (Grimmer, Knox, et al., 2024), Democrats clearly performed better in 2020. The victory was characterized by favorable compositional forces (increasing share of non-White voters, particularly Black and Asian voters) combined with positive rate effects among White voters. The RCVD decomposition confirms that mobilization efforts and demographic trends worked synergistically with preference shifts to yield a substantial Democratic gain. This period represents the high-water mark of the conventional wisdom that demographic change inherently favors Democrats. However, even in 2020, warning signs appeared: all non-White groups voted for Democrats at

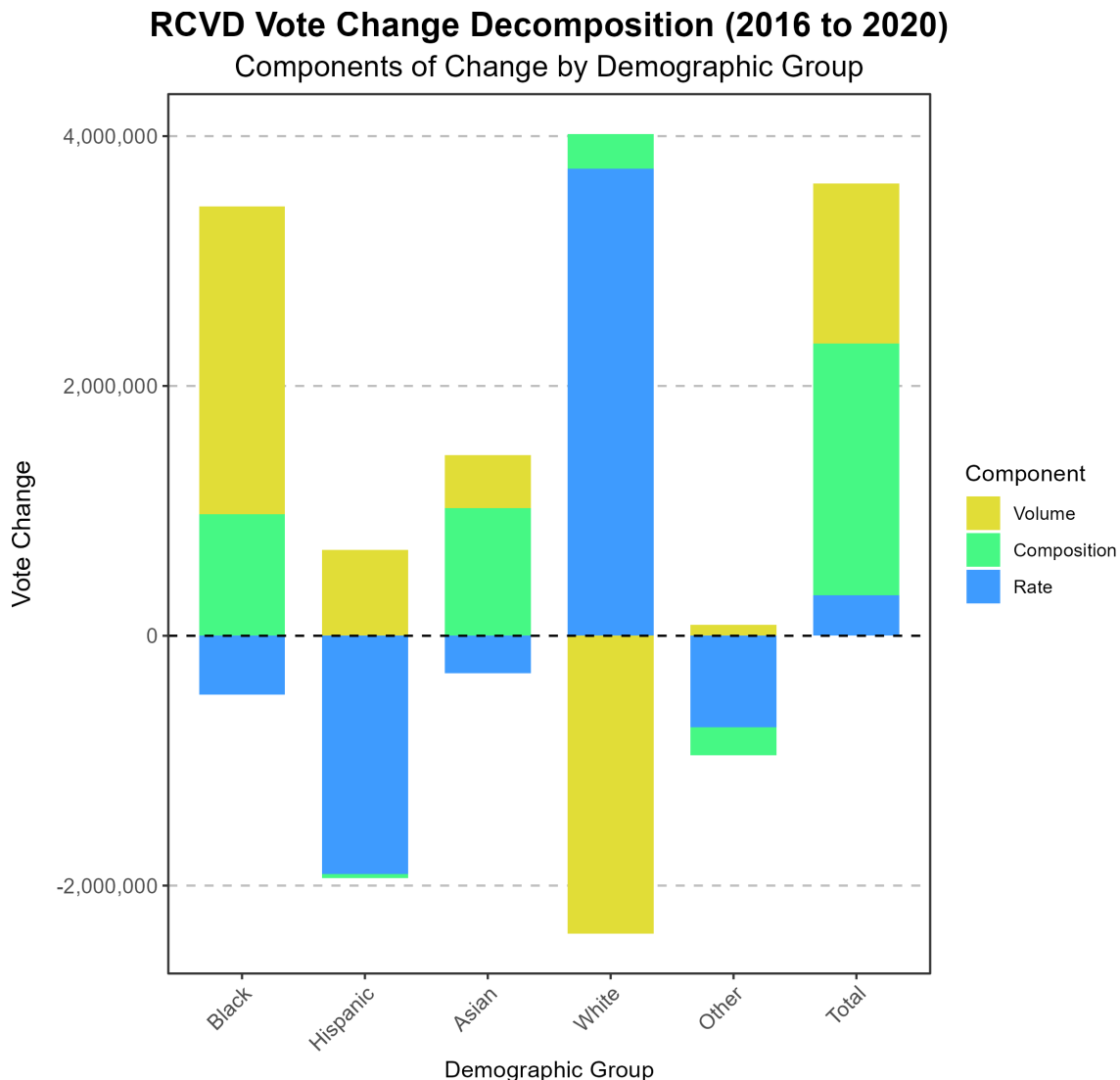


Figure 6: This figure provides an evaluation of the 2020 election compared to the 2016 election using Pew data and the RCVD approach. It shows that composition had a large positive impact for Democrats as a result of shifting out of White voters and into Black and Asian voters. There was also a large positive impact for Democrats driven by improved margins among White voters. However, this impact was almost entirely offset by declining Democratic margins among Hispanic voters, and to a lesser extent among Black, Asian and other voters.

lower rates, with Democratic vote share among Hispanic voters declining noticeably (from 72% to 61%, a margin shift from +44% to +22%).

The Critical 2020 to 2024 Swing— Figure 7

The 2020 to 2024 period represents a critical realignment that challenges established assumptions about the stability of the non-White Democratic coalition. The RCVD analysis reveals a dramatic reversal: Democrats suffered a severe net loss attributable

overwhelmingly to the rate component across all non-White voting groups.

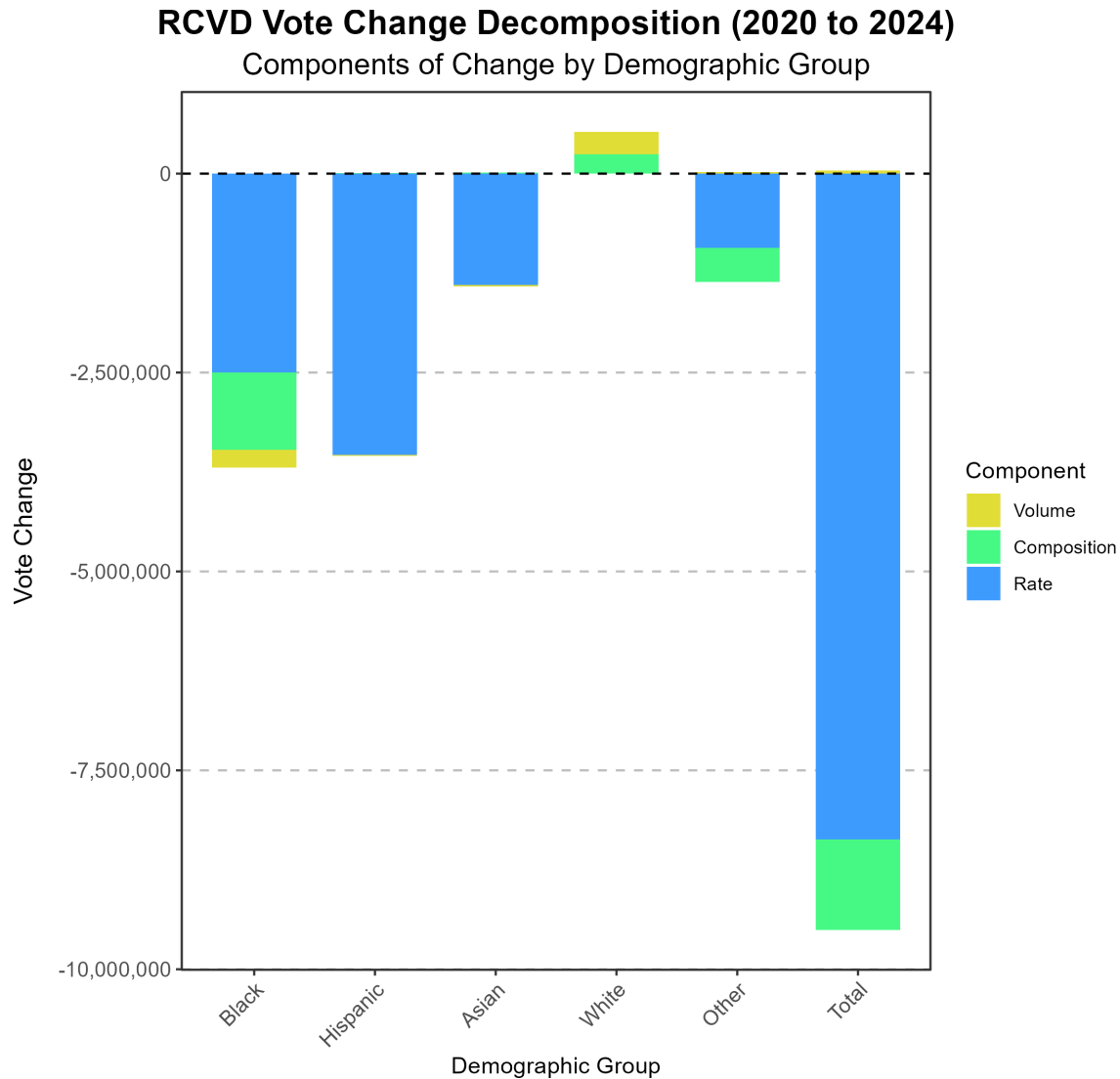


Figure 7: This figure provides an analysis of the 2024 election compared to the 2020 election. It shows that Democrats suffered a dramatic decrease in vote margins among all non-White voting groups, most prominently among Hispanic voters. The decline in support was so significant that even with continued shifts away from White voters amongst the composition of the electorate, the total impact of composition in 2024 was a net negative for Democrats.

The rate effect was negative and substantial among Hispanic and Black voters, signaling erosion of Democratic loyalty and successful Republican persuasion. This negative rate swing was so significant that, even with continued shifts away from White voters in the electorate's composition, the total composition effect in 2024 was net negative for Democrats. This finding underscores the core mechanism of the 2024 result: declining base loyalty (rate) entirely neutralized and then overcame the struc-

tural advantages of demographic change (composition).

The Cumulative 2016 to 2024 Arc—Figure 8

The eight-year view from 2016 to 2024 provides the final synthesis. While the analysis demonstrates some favorable impacts from composition over the long run, this benefit was dwarfed by the cumulative negative trend in the rate component.

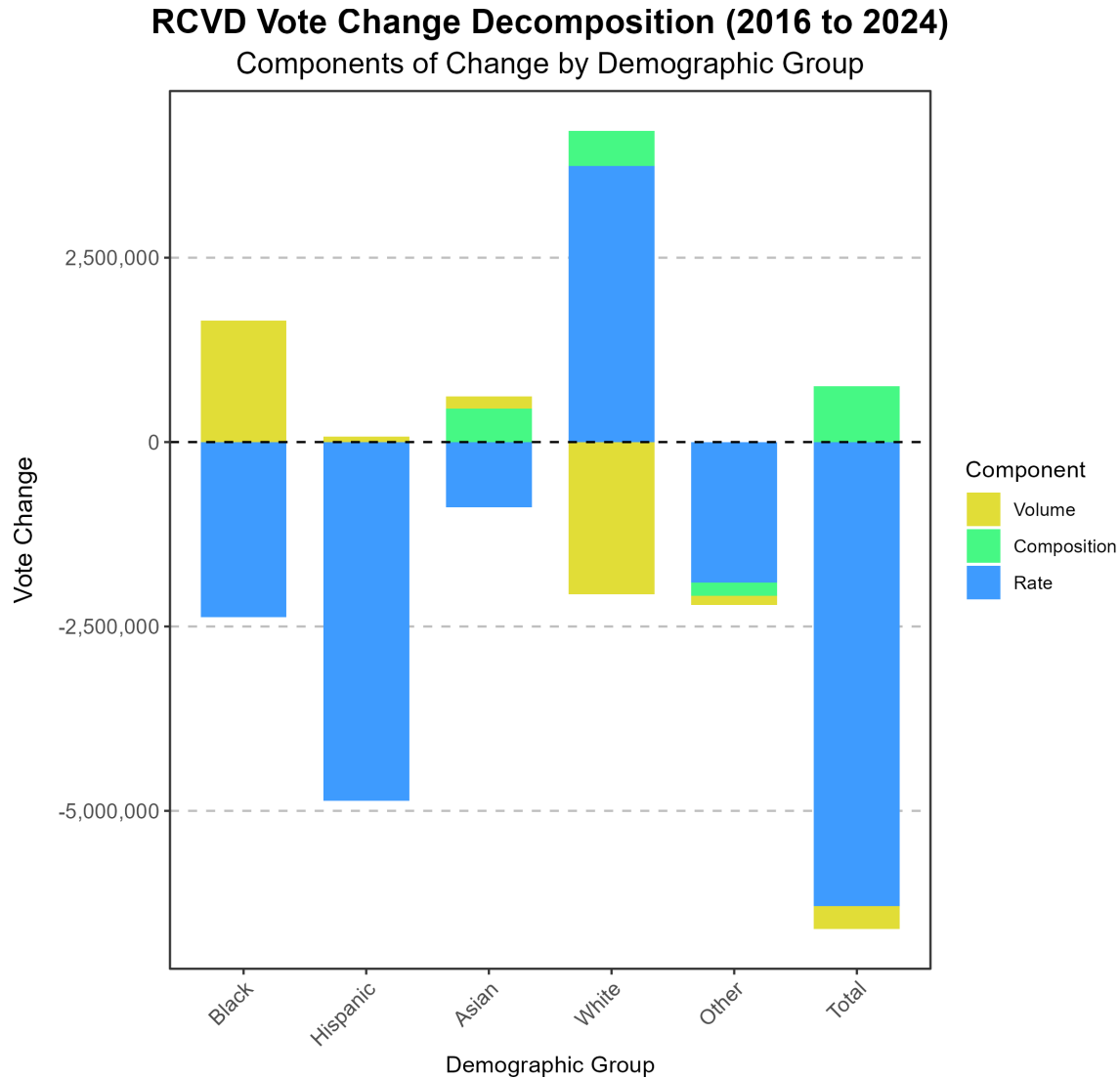


Figure 8: This figure compares the 2024 election to the 2016 election. It highlights the strong negative trend in Democratic vote margins among Hispanic voters, and to a lesser extent among other non-White voters. While this analysis demonstrates some favorable impacts from composition, it highlights the shortcomings of a reliance on these demographic shifts to promote future success for the Democratic party.

The negative rate trend among Hispanic voters is the single most important factor

identified by the RCVD framework over this period. While Democrats showed positive growth among White voters, the framework cleanly partitions the positive effect of an increasing non-White share from the negative effect of these groups becoming less reliably Democratic. The final outcome is explained not by failure of the electorate to grow (volume) or shift demographically (composition), but by failure of the Democratic Party to retain the loyalty of the electorate that did grow (rate).

The rate component identified by RCVD is consistent with recent evidence on the mechanisms behind Latinos’ pro-GOP shift. **hopkins_marble_2026** finds that between 2016 and 2024, Latinos became significantly more restrictionist on immigration while the weight they placed on immigration in their vote choice remained stable. This pattern — changing attitudes with stable issue weights — is precisely what RCVD’s rate component captures: a shift in how a group votes, not a change in who comprises the group or how much weight they assign to particular issues.

The application of RCVD across these three cycles demonstrates that the 2024 result was driven by a decisive decoupling of composition and rate effects. The long-predicted demographic dividend (composition) was neutralized by a collapse of partisan preference (Rate) within the fastest-growing segments of the electorate, a phenomenon masked by aggregate analysis and only revealed by the Rate, Composition, and Volume Decomposition.

5 Conclusion

As demographic shifts reshape the U.S. voting landscape, understanding how changes between groups affect election outcomes is crucial for both scholars and policymakers. Previous methods for explaining these shifts have introduced significant interpretive errors, frequently amounting to millions of misallocated votes, due to the conflation of preference shifts with compositional change. The Rate, Composition, and Volume Decomposition (RCVD) approach, as defined and applied in this paper, resolves this fundamental ambiguity. It provides a zero-loss methodology that precisely specifies how changes in partisan preference (rate), the relative size of groups (composition),

and overall participation (volume) influence electoral outcomes. The RCVD framework is broadly applicable, extending beyond U.S. elections to any context where subdividing populations into distinct groups helps explain dynamics in time-series cross-sectional data.

My empirical application of RCVD to the 2016, 2020, and 2024 presidential elections yields a core finding: the 2024 outcome was driven not by the expected long-term compositional benefit for the Democratic Party, but by a decisive decoupling of composition and rate effects. The framework isolates a substantial negative rate effect—the loss of partisan loyalty—among the fastest-growing non-White electorates, particularly Hispanic voters. This preference volatility neutralized the structural compositional advantage, confirming that electoral success depends less on passive demographic shifts and more on active defense of coalition loyalty. The RCVD framework is thus crucially diagnostic, directing attention from organizational success (volume and composition) toward persuasion challenges (rate).

RCVD’s zero-residual property eliminates decomposition error but does not eliminate measurement error. The framework is only as good as its inputs — particularly the survey-based estimates of group-level vote margins that determine the rate component. As the ANES versus Pew comparison in Section 4 demonstrates, different surveys can produce fundamentally different rate estimates, leading to divergent substantive conclusions about the same election. Researchers applying RCVD should therefore attend carefully to data quality, favoring probability-based surveys with demonstrated accuracy for the subpopulations of interest (Mercer and Lau, 2023; D. J. Hopkins et al., 2024). Notably, RCVD’s exact accounting makes data quality more visible: because there is no residual to absorb measurement error, any discrepancy between the decomposition’s implied vote totals and actual results is attributable entirely to the input data.

Future work will explore the robustness and granularity of the RCVD approach in two key areas. First, I will investigate its ability to account for nested subgroups, demonstrating how shifts within smaller partitions—such as regional or age-based splits

within the Hispanic electorate contribute to broader electoral changes, building on the work of Fraga, Velez, et al. (2024). Second, I will extend the analysis across multiple non-presidential time periods to capture long-term trends, showing how RCVD can reveal the evolving durability and fragility of group shifts over time. These extensions will further validate RCVD as a tool for analyzing historical shifts in political support and for understanding electorate dynamics in future elections.

References

- Achen, Christopher H. and Larry M. Bartels (2016). *Democracy for realists: why elections do not produce responsive government*. Princeton Studies in Political Behavior. Princeton: Princeton University Press. ISBN: 9780691169446.
- ANES (2021). *American National Election Studies*. ANES 2020 Time Series Study Full Release [dataset and documentation]. February 10, 2022 version.
- Autor, David et al. (Oct. 2020). “Importing Political Polarization? The Electoral Consequences of Rising Trade Exposure”. en. In: *American Economic Review* 110.10, pp. 3139–3183. ISSN: 0002-8282. DOI: [10.1257/aer.20170011](https://pubs.aeaweb.org/doi/10.1257/aer.20170011). URL: <https://pubs.aeaweb.org/doi/10.1257/aer.20170011> (visited on 12/17/2024).
- Axelrod, Robert (Mar. 1972). “Where the Votes Come From: An Analysis of Electoral Coalitions, 1952–1968”. en. In: *American Political Science Review* 66.1, pp. 11–20. ISSN: 0003-0554, 1537-5943. DOI: [10.2307/1959275](https://www.cambridge.org/core/product/identifier/S0003055400138882/type/journal_article). URL: https://www.cambridge.org/core/product/identifier/S0003055400138882/type/journal_article (visited on 03/12/2024).
- Bump, Philip (May 2021). “Trump’s improvement with non-White voters in 2020 was important — but not as important as the turnout surge”. In: *The Washington Post*. URL: https://www.washingtonpost.com/politics/2021/05/10/trumps-improvement-with-non-white-voters-2020-was-important-not-important-turnout-surge/?_pml=1 (visited on 03/26/2026).
- Calvo, Richard, Vincent Pons, and Jesse Shapiro (Oct. 2024). *Pitfalls of Demographic Forecasts of US Elections*. en. Tech. rep. w33016. Cambridge, MA: National Bureau of Economic Research, w33016. DOI: [10.3386/w33016](https://www.nber.org/papers/w33016). URL: <http://www.nber.org/papers/w33016.pdf> (visited on 10/16/2025).

- Campbell, Angus, ed. (1960). *The American voter*. eng. 2. Aufl. Chicago, Ill.: Univ. of Chicago Press. ISBN: 978-0-226-09253-9.
- Carmines, Edward G., Michael J. Ensley, and Michael W. Wagner (Dec. 2016). “Ideological Heterogeneity and the Rise of Donald Trump”. en. In: *The Forum* 14.4, pp. 385–397. ISSN: 1540-8884. DOI: [10.1515/for-2016-0036](https://doi.org/10.1515/for-2016-0036). URL: <https://www.degruyter.com/document/doi/10.1515/for-2016-0036/html?lang=en> (visited on 03/12/2024).
- Catalist (2025). “What Happened in 2024”. In: URL: <https://catalist.us/whathappened2024/> (visited on 04/03/2026).
- Citrin, Jack, Eric Schickler, and John Sides (Jan. 2003). “What if Everyone Voted? Simulating the Impact of Increased Turnout in Senate Elections”. en. In: *American Journal of Political Science* 47.1, pp. 75–90. ISSN: 0092-5853, 1540-5907. DOI: [10.1111/1540-5907.00006](https://doi.org/10.1111/1540-5907.00006). URL: <https://onlinelibrary.wiley.com/doi/10.1111/1540-5907.00006> (visited on 10/13/2024).
- Corral, Álvaro J and David L Leal (Aug. 2024). “El Cuento del Destino: Latino Voters, Demographic Determinism, and the Myth of an Inevitable Democratic Party Majority”. en. In: *Political Science Quarterly* 139.3, pp. 335–359. ISSN: 0032-3195, 1538-165X. DOI: [10.1093/psquar/qqae005](https://doi.org/10.1093/psquar/qqae005). URL: <https://academic.oup.com/psq/article/139/3/335/7629974> (visited on 03/30/2026).
- Engelhardt, Andrew M. (July 2019). “Trumped by Race: Explanations for Race’s Influence on Whites’ Votes in 2016”. In: *Quarterly Journal of Political Science* 14.3, pp. 313–328. ISSN: 15540634. DOI: [10.1561/100.00018068](https://doi.org/10.1561/100.00018068). URL: <http://www.nowpublishers.com/article/Details/QJPS-18068> (visited on 03/12/2024).
- Equis Research (2024). “A Preliminary Look at the 2024 Latino Vote”. In: URL: <https://www.weareequis.us/research/prelimlatinovote2024>.

- Fraga, Bernard (Jan. 2016). “Candidates or Districts? Reevaluating the Role of Race in Voter Turnout”. en. In: *American Journal of Political Science* 60.1, pp. 97–122. ISSN: 0092-5853, 1540-5907. DOI: [10.1111/ajps.12172](https://doi.org/10.1111/ajps.12172). URL: <https://onlinelibrary.wiley.com/doi/10.1111/ajps.12172> (visited on 10/31/2025).
- Fraga, Bernard, Sean McElwee, et al. (Dec. 2021). “Analysis | Why did Trump win? More whites — and fewer blacks — actually voted.” en-US. In: *Washington Post*. ISSN: 0190-8286. URL: <https://www.washingtonpost.com/news/monkey-cage/wp/2017/05/08/why-did-trump-win-more-whites-and-fewer-blacks-than-normal-actually-voted/> (visited on 03/12/2024).
- Fraga, Bernard, Yamil R. Velez, and Emily A. West (May 2024). “Reversion to the Mean, or Their Version of the Dream? Latino Voting in an Age of Populism”. en. In: *American Political Science Review*, pp. 1–9. ISSN: 0003-0554, 1537-5943. DOI: [10.1017/S0003055424000406](https://doi.org/10.1017/S0003055424000406). URL: https://www.cambridge.org/core/product/identifier/S0003055424000406/type/journal_article (visited on 12/08/2024).
- Grimmer, Justin, Dean Knox, and Sean Westwood (Aug. 2024). *Assessing the Reliability of Probabilistic US Presidential Election Forecasts May Take Decades*. DOI: [10.31219/osf.io/6g5zq](https://doi.org/10.31219/osf.io/6g5zq). URL: <https://osf.io/6g5zq> (visited on 12/17/2024).
- Grimmer, Justin, William Marble, and Cole Tanigawa-Lau (June 2022). *Measuring the Contribution of Voting Blocs to Election Outcomes*. preprint. SocArXiv. DOI: [10.31235/osf.io/c9fkg](https://doi.org/10.31235/osf.io/c9fkg). URL: <https://osf.io/c9fkg> (visited on 02/18/2024).
- Hartig, Hannah et al. (July 2023). *Republican Gains in 2022 Midterms Driven Mostly by Turnout Advantage*. Tech. rep. Pew Research Center. URL: <https://www.pewresearch.org/politics/2023/07/12/republican-gains->

- [in-2022-midterms-driven-mostly-by-turnout-advantage/](#) (visited on 07/03/2025).
- Hill, Seth J., Daniel J. Hopkins, and Gregory A. Huber (Apr. 2021). “Not by turnout alone: Measuring the sources of electoral change, 2012 to 2016”. en. In: *Science Advances* 7.17, eabe3272. ISSN: 2375-2548. DOI: [10.1126/sciadv.abe3272](#). URL: <https://www.science.org/doi/10.1126/sciadv.abe3272> (visited on 02/18/2024).
- Hopkins, Daniel and William Marble (Mar. 2026). *What Role Did Immigration Attitudes Play in Latinos’ 2016-2024 Swing Toward the GOP?* DOI: [10.31235/osf.io/jgmqh_v2](#). URL: https://osf.io/jgmqh_v2 (visited on 03/30/2026).
- Hopkins, Daniel J. et al. (Nov. 2024). “Getting the Race Wrong: A Case Study of Sampling Bias and Black Voters in Online, Opt-In Polls”. en. In: *The Journal of Race, Ethnicity, and Politics* 9.3, pp. 429–441. ISSN: 2056-6085. DOI: [10.1017/rep.2024.11](#). URL: https://www.cambridge.org/core/product/identifier/S2056608524000114/type/journal_article (visited on 03/30/2026).
- Klein, Ezra (Jan. 2024). *The Ezra Klein Show: ‘The Strongest Democratic Party That Any of Us Have Ever Seen’ on Apple Podcasts*. en-US. URL: <https://podcasts.apple.com/us/podcast/the-strongest-democratic-party-that-any-of-us-have-ever-seen/id1548604447?i=1000642890587> (visited on 05/08/2024).
- Lemi, Danielle Casarez (Dec. 2021). “Do Voters Prefer Just Any Descriptive Representative? The Case of Multiracial Candidates”. en. In: *Perspectives on Politics* 19.4, pp. 1061–1081. ISSN: 1537-5927, 1541-0986. DOI: [10.1017/S1537592720001280](#). URL: https://www.cambridge.org/core/product/identifier/S1537592720001280/type/journal_article (visited on 09/30/2024).

- Lopez, Mark Hugo and Paul Taylor (2009). *Dissecting the 2008 Electorate: Most Diverse in U.S. History*. Tech. rep. Pew Research Center. URL: <https://www.pewresearch.org/race-and-ethnicity/2009/04/30/dissecting-the-2008-election-most-diverse-in-us-history/> (visited on 10/12/2024).
- Marble, William, Justin Grimmer, and Cole Tanigawa-Lau (Sept. 2024). “Measuring the Contribution of Voting Blocs to Election Outcomes”. en. In: *The Journal of Politics*, p. 732964. ISSN: 0022-3816, 1468-2508. DOI: [10.1086/732964](https://doi.org/10.1086/732964). URL: <https://www.journals.uchicago.edu/doi/10.1086/732964> (visited on 09/30/2024).
- Mercer, Andrew and Arnold Lau (Sept. 2023). “Comparing Two Types of Online Survey Samples”. In: URL: <https://www.pewresearch.org/methods/2023/09/07/comparing-two-types-of-online-survey-samples/> (visited on 03/30/2026).
- Reed, Michael (1998). *Fundamental ideas of analysis*. eng. New York Chichester: John Wiley. ISBN: 978-0-471-15996-4.
- Republican National Committee (2013). “Growth and Opportunity Project”. In: URL: <https://www.wsj.com/public/resources/documents/RNCreport03182013.pdf> (visited on 04/03/2026).
- Sides, John, Michael Tesler, and Lynn Vavreck (Dec. 2019). *Identity Crisis: The 2016 Presidential Campaign and the Battle for the Meaning of America*. Princeton University Press. ISBN: 978-1-4008-8830-6. DOI: [10.1515/9781400888306](https://doi.org/10.1515/9781400888306). URL: <https://www.degruyter.com/document/doi/10.1515/9781400888306/html> (visited on 10/13/2024).
- U.S. Census Bureau (2024). *American Community Survey 1-Year Estimates, Table B03003: Hispanic or Latino Origin*. URL: <https://data.census.gov/> (visited on 03/26/2026).

- Wooldridge, Jeffrey M. (2010). *Econometric Analysis of Cross Section and Panel Data*. The MIT Press. ISBN: 9780262232586. URL: <https://www.jstor.org/stable/j.ctt5hhcfr> (visited on 03/20/2024).
- Wooley, John and Gerhard Peters (Dec. 2024). “The American Presidency Project.” In: URL: <https://www.presidency.ucsb.edu/statistics/elections/2024> (visited on 10/31/2025).
- Woolley, John and Gerhard Peters (2021). *Voter Turnout in Presidential Elections*. URL: <https://www.presidency.ucsb.edu/statistics/data/voter-turnout-in-presidential-elections> (visited on 10/12/2024).
- Zingher, Joshua N. (Aug. 2019). “An analysis of the changing social bases of America’s political parties: Group support in the 2012 and 2016 presidential elections”. en. In: *Electoral Studies* 60, p. 102042. ISSN: 02613794. DOI: [10.1016/j.electstud.2019.04.006](https://doi.org/10.1016/j.electstud.2019.04.006). URL: <https://linkinghub.elsevier.com/retrieve/pii/S0261379419300253> (visited on 02/18/2024).

Appendix

On Overfitting

While zero-error terms in regression models often raise concerns about overfitting (Wooldridge, 2010), the RCVD approach is not subject to these risks. As it is not a regression or estimation technique, but rather an accounting framework, zero-error reflects the accurate decomposition of shifts rather than problematic estimation.

Proof of Equation 2

$$\begin{aligned}\Delta z(t) &= z(t_2) - z(t_1) \\ &= r(t_2)c(t_2) - r(t_1)c(t_1) \\ r(t_2)c(t_2) - r(t_1)c(t_1) &= r(t_2)c(t_2) + r(t_1)c(t_1) - r(t_1)c(t_1) + r(t_2)c(t_1) - r(t_2)c(t_1) + r(t_1)c(t_2) - r(t_1)c(t_2) - r(t_1)c(t_1) \\ &= [r(t_2)c(t_1) - r(t_1)c(t_1)] + [r(t_1)c(t_2) - r(t_1)c(t_1)] + [r(t_2)c(t_2) - r(t_1)c(t_2) - r(t_2)c(t_1) + r(t_1)c(t_1)] \\ r(t_2)c(t_2) - r(t_1)c(t_1) &= [r(t_2) - r(t_1)]c(t_1) + r(t_1)[c(t_2) - c(t_1)] + [r(t_2) - r(t_1)][c(t_2) - c(t_1)]\end{aligned}$$

Proof of Equation 3

$$\begin{aligned}\Delta z(t) &= z(t_2) - z(t_1) \\ &= r(t_2)c(t_2)v(t_2) - r(t_1)c(t_1)v(t_1) \\ r(t_2)c(t_2)v(t_2) - r(t_1)c(t_1)v(t_1) &= r(t_2)c(t_2)v(t_2) - r(t_2)c(t_1)v(t_1) + r(t_2)c(t_1)v(t_1) \\ &\quad - r(t_1)c(t_2)v(t_1) + r(t_1)c(t_2)v(t_1) - r(t_1)c(t_1)v(t_2) + r(t_1)c(t_1)v(t_2) \\ &\quad - r(t_2)c(t_2)v(t_1) + r(t_2)c(t_2)v(t_1) - r(t_2)c(t_1)v(t_2) + r(t_2)c(t_1)v(t_2) \\ &\quad - r(t_1)c(t_2)v(t_2) + r(t_1)c(t_2)v(t_2) - r(t_1)c(t_1)v(t_1) \\ &= 3r(t_1)c(t_1)v(t_1) - 3r(t_1)c(t_1)v(t_1) \\ &\quad + r(t_2)c(t_1)v(t_1) - 2r(t_2)c(t_1)v(t_1) \\ &\quad + r(t_1)c(t_2)v(t_1) - 2r(t_1)c(t_2)v(t_1) \\ &\quad + r(t_1)c(t_1)v(t_2) - 2r(t_1)c(t_1)v(t_2) \\ &\quad + r(t_2)c(t_2)v(t_1) + r(t_2)c(t_1)v(t_2) + r(t_1)c(t_2)v(t_2) \\ &\quad + [r(t_2)c(t_2)v(t_2) - r(t_2)c(t_2)v(t_1) - r(t_2)c(t_1)v(t_2) + r(t_2)c(t_1)v(t_1)] \\ &\quad - r(t_1)c(t_2)v(t_2) + r(t_1)c(t_1)v(t_2) + r(t_1)c(t_2)v(t_1) - r(t_1)c(t_1)v(t_1)] \\ &= [r(t_2)c(t_1)v(t_1) - r(t_1)c(t_1)v(t_1)] \\ &\quad + [r(t_1)c(t_2)v(t_1) - r(t_1)c(t_1)v(t_1)]\end{aligned}$$

$$\begin{aligned}
& + [r(t_1)c(t_1)v(t_2) - r(t_1)c(t_1)v(t_1)] \\
& + [r(t_2)c(t_2) - r(t_2)c(t_1) - r(t_1)c(t_2) + r(t_1)c(t_1)]v(t_1) \\
& + [r(t_2)v(t_2) - r(t_2)v(t_1) - r(t_1)v(t_2) + r(t_1)v(t_1)]c(t_1) \\
& + [c(t_2)v(t_2) - c(t_2)v(t_1) - c(t_1)v(t_2) + c(t_1)v(t_1)]r(t_1) \\
& + [r(t_2)c(t_2)v(t_2) - r(t_2)c(t_2)v(t_1) - r(t_2)c(t_1)v(t_2) + r(t_2)c(t_1)v(t_1) \\
& - r(t_1)c(t_2)v(t_2) + r(t_1)c(t_1)v(t_2) + r(t_1)c(t_2)v(t_1) - r(t_1)c(t_1)v(t_1)] \\
r(t_2)c(t_2)v(t_2) - r(t_1)c(t_1)v(t_1) & = [r(t_2) - r(t_1)]c(t_1)v(t_1) + r(t_1)[c(t_2) - c(t_1)]v(t_1) + r(t_1)c(t_1)[v(t_2) - v(t_1)] \\
& + [r(t_2) - r(t_1)][c(t_2) - c(t_1)]v(t_1) + [r(t_2) - r(t_1)]c(t_1)[v(t_2) - v(t_1)] + r(t_1)[c(t_2) - c(t_1)][v(t_2) - v(t_1)] \\
& + [r(t_2) - r(t_1)][c(t_2) - c(t_1)][v(t_2) - v(t_1)]
\end{aligned}$$

Proof of Equation 3 with Description and Color Coding:

$$\begin{aligned}
\Delta z(t) & = z(t_2) - z(t_1) \\
& = r(t_2)c(t_2)v(t_2) - r(t_1)c(t_1)v(t_1)
\end{aligned}$$

The core expansion begins here. I begin by color coding terms which are introduced that cancel

$$\begin{aligned}
r(t_2)c(t_2)v(t_2) - r(t_1)c(t_1)v(t_1) & = r(t_2)c(t_2)v(t_2) + 2[r(t_2)c(t_1)v(t_1) - r(t_2)c(t_1)v(t_1)] \\
& + 2[r(t_1)c(t_2)v(t_1) - r(t_1)c(t_2)v(t_1)] + 2[r(t_1)c(t_1)v(t_2) - r(t_1)c(t_1)v(t_2)] \\
& + [r(t_2)c(t_2)v(t_1) - r(t_2)c(t_2)v(t_1)] + [r(t_2)c(t_1)v(t_2) - r(t_2)c(t_1)v(t_2)] \\
& + [r(t_1)c(t_2)v(t_2) - r(t_1)c(t_2)v(t_2)] + 3[r(t_1)c(t_1)v(t_1) - r(t_1)c(t_1)v(t_1)] \\
& - r(t_1)c(t_1)v(t_1)
\end{aligned}$$

After rearranging, I drop colors for terms that will be completely reconfigured, only maintaining color groupings for terms that are essentially finalized

$$\begin{aligned}
& = 3r(t_1)c(t_1)v(t_1) - 3r(t_1)c(t_1)v(t_1) \\
& + r(t_2)c(t_1)v(t_1) - 2r(t_2)c(t_1)v(t_1) \\
& + r(t_1)c(t_2)v(t_1) - 2r(t_1)c(t_2)v(t_1) \\
& + r(t_1)c(t_1)v(t_2) - 2r(t_1)c(t_1)v(t_2) \\
& + r(t_2)c(t_2)v(t_1) + r(t_2)c(t_1)v(t_2) + r(t_1)c(t_2)v(t_2) \\
& + [r(t_2)c(t_2)v(t_2) - r(t_2)c(t_2)v(t_1) - r(t_2)c(t_1)v(t_2) + r(t_2)c(t_1)v(t_1) \\
& - r(t_1)c(t_2)v(t_2) + r(t_1)c(t_1)v(t_2) + r(t_1)c(t_2)v(t_1) - r(t_1)c(t_1)v(t_1)]
\end{aligned}$$

After additional rearranging, I apply color groupings for terms as they will appear in the final proof

$$\begin{aligned}
&= [r(t_2)c(t_1)v(t_1) - r(t_1)c(t_1)v(t_1)] \\
&+ [r(t_1)c(t_2)v(t_1) - r(t_1)c(t_1)v(t_1)] \\
&+ [r(t_1)c(t_1)v(t_2) - r(t_1)c(t_1)v(t_1)] \\
&+ [r(t_2)c(t_2) - r(t_2)c(t_1) - r(t_1)c(t_2) + r(t_1)c(t_1)]v(t_1) \\
&+ [r(t_2)v(t_2) - r(t_2)v(t_1) - r(t_1)v(t_2) + r(t_1)v(t_1)]c(t_1) \\
&+ [c(t_2)v(t_2) - c(t_2)v(t_1) - c(t_1)v(t_2) + c(t_1)v(t_1)]r(t_1) \\
&+ [r(t_2)c(t_2)v(t_2) - r(t_2)c(t_2)v(t_1) - r(t_2)c(t_1)v(t_2) + r(t_2)c(t_1)v(t_1) \\
&\quad - r(t_1)c(t_2)v(t_2) + r(t_1)c(t_1)v(t_2) + r(t_1)c(t_2)v(t_1) - r(t_1)c(t_1)v(t_1)] \\
r(t_2)c(t_2)v(t_2) - r(t_1)c(t_1)v(t_1) &= [r(t_2) - r(t_1)]c(t_1)v(t_1) + r(t_1)[c(t_2) - c(t_1)]v(t_1) + r(t_1)c(t_1)[v(t_2) - v(t_1)] \\
&+ [r(t_2) - r(t_1)][c(t_2) - c(t_1)]v(t_1) + [r(t_2) - r(t_1)]c(t_1)[v(t_2) - v(t_1)] \\
&+ r(t_1)[c(t_2) - c(t_1)][v(t_2) - v(t_1)] \\
&+ [r(t_2) - r(t_1)][c(t_2) - c(t_1)][v(t_2) - v(t_1)] \Rightarrow \\
\Delta z(t) &= \Delta r(t)c(t_1)v(t_1) + r(t_1)\Delta r(t)v(t_1) + r(t_1)c(t_1)\Delta v(t) \\
&+ \Delta r(t)\Delta c(t)v(t_1) + \Delta r(t)c(t_1)\Delta v(t) \\
&+ r(t_1)\Delta c(t)\Delta v(t) \\
&+ \Delta r(t)\Delta c(t)\Delta v(t)
\end{aligned}$$

Proof of Key Equation 5

$$\begin{aligned}
\Delta z(t) &= z(t_2) - z(t_1) \\
z(t_2) - z(t_1) &= r(t_2)c(t_2)v(t_2) - r(t_1)c(t_1)v(t_1) \\
&= r(t_2)[c(t_1)v(t_1) - c(t_1)v(t_1) + c(t_2)v(t_1) - c(t_2)v(t_1) + c(t_2)v(t_2)] - \\
&\quad r(t_1)c(t_1)v(t_1) \\
&= r(t_2)c(t_1)v(t_1) - r(t_1)c(t_1)v(t_1) + r(t_2)c(t_2)v(t_1) - \\
&\quad r(t_2)c(t_1)v(t_1) + r(t_2)c(t_2)v(t_2) - r(t_2)c(t_2)v(t_1) \\
&= [r(t_2) - r(t_1)]c(t_1)v(t_1) + \\
&\quad r(t_2)[c(t_2) - c(t_1)]v(t_1) + \\
&\quad r(t_2)c(t_2)[v(t_2) - v(t_1)]
\end{aligned} \tag{6}$$

Two-Group Simulation: Quantifying the Residual

The simple example in Section 3 used hypothetical data to illustrate the mechanics of the LDD’s residual problem. Applying the same analysis to actual election data reveals that these residuals grow substantially larger in real-world applications. Table A1 presents a simplified analysis of the 2016 to 2020 presidential election, collapsing racial groups into White and non-White voters. The data is derived from the framework introduced by Marble et al. (2024), with vote totals rounded for clarity of presentation.⁵⁶

Table A1 shows the magnitude of the change we must decompose: a total shift of 9 million votes for the Democratic candidate (from Clinton’s +8 million margin to Biden’s +17 million margin). This shift is characterized by changes in all three fundamental parameters. With respect to rate, there was a 7.5% decline in margin within the non-White bloc, countered by a 6.6% improvement in margin within the White bloc. In terms of composition, there was a 3.6% shift of the electorate towards non-White voters. Finally, there was a large increase in volume of 26 million additional total votes cast. Table A2 shows the decomposition.

Figure A1 demonstrates how the LDD fails to fully account for the total 9 million vote change. When applied to this two-group system, the LDD generates a Mean Absolute Error (MAE) of just over 1.4 million votes — roughly 15% of the total 9 million vote change being decomposed. This unallocated portion is the residual term—real votes that contributed to the 2020 outcome but are mathematically unassigned to any specific component (rate, composition, or volume).

As established in Section 2, this non-zero residual creates interpretive problems.

Since previous decomposition studies have left the volume term unspecified, they have,

⁵The Democratic party is consistently set as the reference group across all analyses. Positive numbers indicate increased support/votes for Democrats; negative numbers indicate decreased support. No political endorsement should be inferred from this mathematical designation.

⁶I have replicated the process used by Marble et al. (2024) in creating this data, which is based upon the data from ANES (2021). The difference in their reported votes from reality seems to be related to their estimation of voter support for candidates based on the ANES. Vote totals are rounded to the nearest half-million for ease of presentation; Section 4 provides more precise figures.

Table A1: Comparing Two Elections — Basic Data

Election	Group	Cand. 1	Cand. 2	Total	Margin	Mar. Rate (%)	Comp. (%)
1st Election (2016)	Non-White	29.5	7.5	37.0	+22.0	+59.5	28.7
	White	39.0	53.0	92.0	−14.0	−15.2	71.3
	Total	68.5	60.5	129.0	+8.0	+6.2	100.0
2nd Election (2020)	Non-White	38.0	12.0	50.0	+26.0	+52.0	32.3
	White	48.0	57.0	105.0	−9.0	−8.6	67.7
	Total	86.0	69.0	155.0	+17.0	+11.0	100.0
Comparing the Two Elec- tions	Non-White	8.5	4.5	13.0	+4.0	−7.5	+3.6
	White	9.0	4.0	13.0	+5.0	+6.6	−3.6
	Total	17.5	8.5	26.0	+9.0	+4.8	0.0

Votes are reported in millions. Candidate 1 refers to Clinton (2016) and Biden (2020); Candidate 2 refers to Trump.

Table A2: Concept Demonstration

Method	Group	Rate	Comp.	Volume	Calc Mar Chng	Var to Act	Var. %
Derivative Calculation	Non-White	−2.76	2.74	4.43	4.42	+0.42	+10.4
	White	+6.11	0.70	−2.82	3.99	−1.01	−20.1
	Total	+3.35	3.44	1.61	8.41	−0.59	−6.5
RCV Calculation	Non-White	−2.76	2.40	4.36	4.00	0.00	0.0
	White	+6.11	0.40	−1.51	5.00	0.00	0.0
	Total	+3.35	2.79	2.85	9.00	0.00	0.0
Comparing the Two Ap- proaches	Non-White		+0.34	+0.07	+0.42		
	White		+0.31	−1.31	−1.01		
	Total		+0.65	−1.24	−0.59		

All

numbers are in millions. Variance percentages are shown with one decimal place.

in essence, folded the residual into an error term that also contained the correctly calculated volume effect. As demonstrated in Figure A2, this leads to systematic misattribution. The benefit Biden received from the change in the racial composition of the electorate is understated by 650,000 votes using the LDD, while a large portion of the 1.4 million residual is incorrectly assigned, masking the true contribution of the volume effect. The residual manifests primarily as an overstatement of the compositional effect and an understatement of the volume effect — the LDD credits composition with votes that should be attributed to the overall expansion of the electorate.



Figure A1: This figure shows that even in a simple case with only two voter groups, the LDD calculation can yield misattributed votes on an order of magnitude of over one million votes.

By contrast, the RCVD approach leaves a MAE of 0, precisely accounting for every vote in the system. The RCVD approach retains the highly intuitive property of explaining rate in a manner identical to the LDD, meaning that the large body of existing work focused solely on rate changes remains valid, while the problematic composition and volume terms are corrected.⁷⁸

The Problem is Not Volume: Isolating Rate-Composition Misspecification

In the preceding analysis, the LDD produced a significant residual, partially driven by unassigned interaction effects and partially by the unmodeled volume term. The traditional literature often assumes this error is synonymous with unexplainable variance or

⁷See the Robustness subsection below for the robustness of the method to partitions.

⁸A natural question is whether the LDD's residual might simply result from omitting the volume component rather than from mismodeling interaction terms. The Problem is Not Volume subsection below addresses this directly by analyzing a controlled scenario where volume is held constant at zero. Even with no volume change, the LDD still produces a residual of 1.7 million votes, definitively demonstrating that the residual stems from the interaction term problem, not from volume omission.

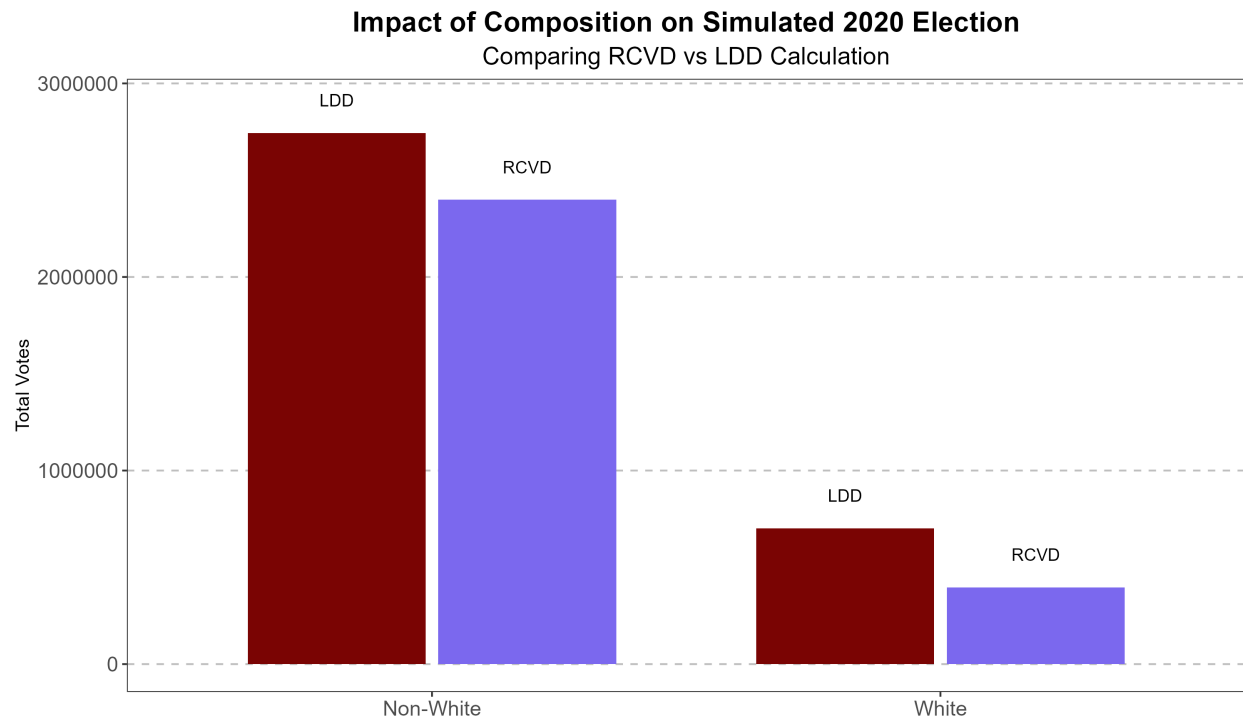


Figure A2: This figure shows that even in a simple case with only two voter groups, the LDD calculation can misattribute the impact of shifts in composition on an order of magnitude of hundreds of thousands of votes.

volume. In this section, I directly challenge that assumption by demonstrating that the LDD’s fundamental flaw lies in its handling of the rate and composition interaction, even when Volume is perfectly controlled.

To achieve this isolation, I analyze a hypothetical, zero-sum vote shift that transforms the actual 2020 Biden popular vote victory into a narrow Trump victory. I ensure this shift is achieved by manipulating only the rate (group vote loyalty) and composition (group size) parameters, while holding the total Volume of votes constant. Since changes in volume can be modeled as a scaling factor, proportionally adjusting the total while preserving the underlying relationships, controlling volume to zero ensures that any resulting error must be driven exclusively by the misspecification of the rate and composition interaction.

Table A3 presents this hypothetical scenario, utilizing a five-group breakdown of the electorate based on data from Marble et al. (2024). I achieve the synthetic shift by multiplying the rate of support for the Democratic candidate by a factor of .93 and

shifting the composition of all non-White groups by a factor of .8. The final line of the 'Total' column confirms the successful manipulation: the net change in total votes across the two scenarios is exactly 0.00 million votes. Therefore, in this controlled, synthetic example, the only two theoretical buckets for explaining the resulting ≈ 17.75 million vote margin shift are rate within groups and composition between them.

Table A3: Hypothetical Data, Both Rate and Composition

Election	Group	Cand. 1	Cand. 2	Total	Gross Mar.	Mar. (%)	Comp. (%)	
1st Election (2020)	Black	15.42	1.47	16.89	+13.95	+82.60	10.89	Votes
	Hispanic	13.21	4.47	17.68	+8.73	+49.40	11.41	
	Other	8.64	5.16	13.80	+3.48	+25.20	8.90	
	White	47.95	57.21	105.16	-9.25	-8.80	67.84	
	NA	0.80	0.67	1.47	+0.13	+8.60	0.95	
	Total	86.02	68.98	155.00	+17.03	+10.99	100.00	
2nd Election (2020)	Black	11.47	2.04	13.51	+9.43	+69.82	8.72	
	Hispanic	9.83	4.32	14.14	+5.51	+38.94	9.12	
	Other	6.43	4.61	11.04	+1.81	+16.44	7.12	
	White	48.82	66.30	115.13	-17.48	-15.18	74.28	
	NA	0.60	0.58	1.18	+0.01	+1.00	0.76	
	Total	77.14	77.86	155.00	-0.71	-0.46	100.00	
Comparing the Two Elections	Black	-3.95	0.57	-3.38	-4.52	-12.78	-2.18	
	Hispanic	-3.38	-0.16	-3.54	-3.23	-10.46	-2.28	
	Other	-2.21	-0.55	-2.76	-1.66	-8.76	-1.78	
	White	0.87	9.10	9.97	-8.23	-6.38	+6.43	
	NA	-0.20	-0.09	-0.29	-0.11	-7.60	-0.19	
	Total	-8.87	8.87	0.00	-17.75	-11.45	0.00	

are reported in millions.

Figure A3 highlights the critical methodological result. Despite the total change in votes being zero, the LDD still produces a massive Mean Absolute Error (MAE) of 1.7 million votes. This unallocated error is caused entirely by the LDD's inability to correctly model the non-linear interaction between rate and composition. By comparison, the RCVD framework correctly accounts for all votes in the system, yielding an MAE of 0.

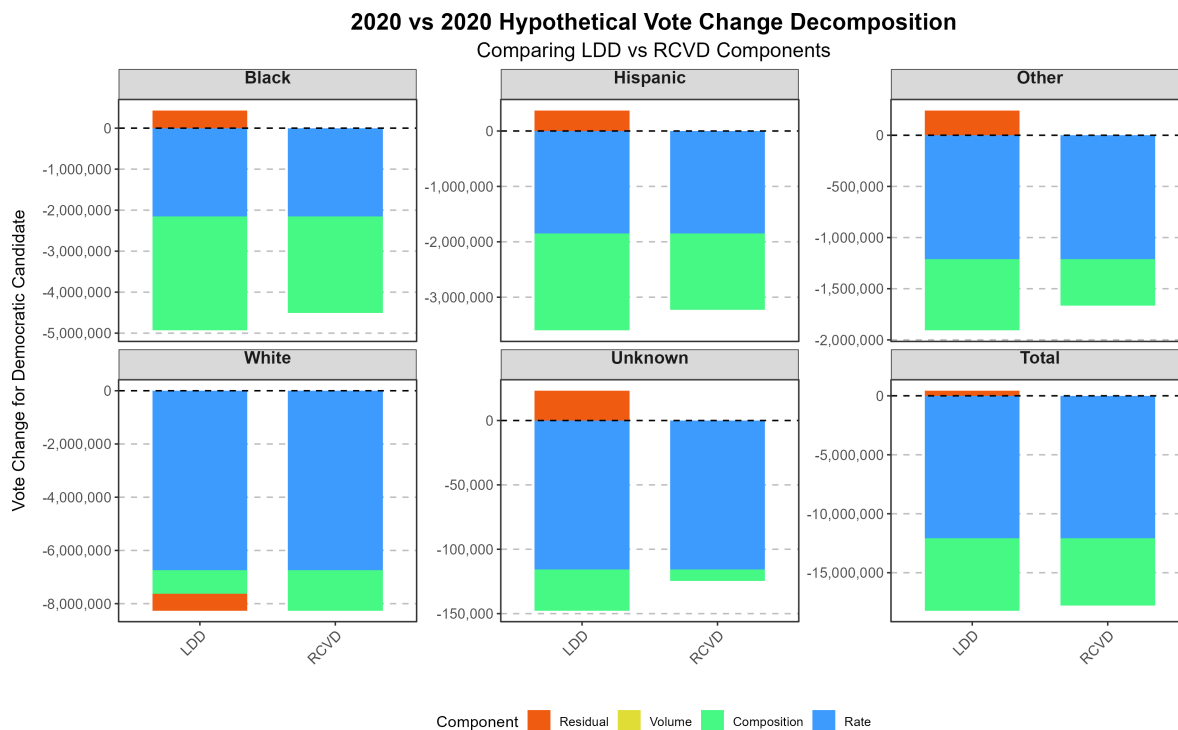


Figure A3: This figure shows that even when volume is not a factor in evaluating the shifts in an election, the LDD creates significant errors in attribution of vote changes. Here, the total absolute error is 1.7 million votes. The underlying numbers reflected in this figure are in Appendix A.

This finding is definitive: The 1.7 million vote error is not due to the omission of volume, but is a direct consequence of the LDD's formula. By comparing the LDD and RCVD, it is clear that all 1.7 million unidentified votes come from a misspecification of the impact of composition and rate, distributed across the groups in increments of hundreds of thousands of votes.

In summary, this scenario definitively refutes the implicit assumption that the LDD's unexplained variance is merely Volume or general error. It validates the necessity of the zero-loss RCVD framework by demonstrating that even in a highly controlled environment where volume is eliminated as a potential explanatory variable, the LDD fundamentally fails to correctly partition the vote change between rate and composition.

Non-Uniqueness and the Fixed RCVD Path

As noted in the conclusion of Section 2, the non-linear nature of the Rate, Composition, and Volume Decomposition (RCVD) means that the order of calculation is not unique. Mathematically, there are six possible permutations of the Rate (ΔR), Composition (ΔC), and Volume (ΔV) terms that satisfy the zero-loss property of the RCVD framework. This path-dependence is a necessary artifact of correctly modeling the interaction effects that the LDD previously relegated to the unassigned residual. Therefore, there needs to be a strong theoretical justification for the chosen order ($\Delta R \rightarrow \Delta C \rightarrow \Delta V$).

Figure A4 illustrates an alternative zero-loss order (composition $\rightarrow$ rate $\rightarrow$ volume) when applied to the simple 2016-2020 simulation. While the total MAE remains 0, the calculated contribution of rate and composition shifts: the alternative order assigns a slightly larger role to composition and a slightly smaller role to rate.

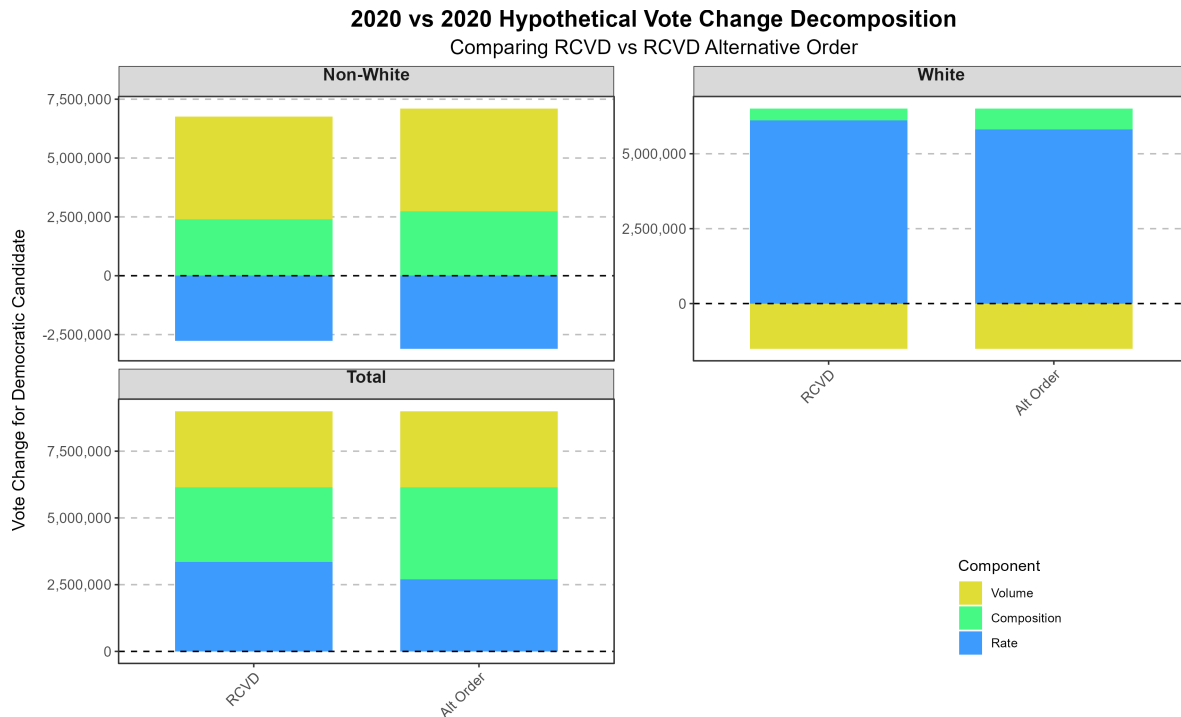


Figure A4: This figure shows that shifting the order in which RCVD is calculated yields no error term, but can shift the interpretation of the impacts of both rate and composition.

Despite the relative closeness of these results in the simple case, I adhere to the

$\Delta R \rightarrow \Delta C \rightarrow \Delta V$ order for two core reasons related to the underlying political theory. First, this order preserves rate’s causal primacy. The convention in electoral studies has always been to calculate the change in group loyalty (rate) based on the prior period’s data. Maintaining this sequence ensures that the ΔR term represents the clean, unadulterated shift in group preference, free from the simultaneous influence of demographic change. This allows our ΔR term to retain the exact same highly intuitive value calculated by the LDD, validating the existing literature while correcting the other terms.

Second, this order maintains volume as a scaling factor. By sequencing volume last, we ensure it serves as a scaling factor—a perfect proportional shift that occurs after all components affecting the margin have been calculated. Intuitively, shifting composition first (e.g., more non-White voters) will ultimately change the final national aggregate rate, even if the rates within each group do not change. By placing ΔC before ΔV , we ensure that volume is purely an increase or decrease in the system’s size, distributed proportionally across the new electoral configuration set by rate and composition. This preserves the most theoretically clean interpretation of volume.

In sum, the $\Delta R \rightarrow \Delta C \rightarrow \Delta V$ ordering is not arbitrary, but a deliberate choice that preserves the most theoretically clean and consistent interpretation of rate against existing literature, while ensuring composition and volume are correctly specified.

Rate Only and Composition Only

Table A4 shows how rates would have had to have shifted in order to guarantee a win for Trump in 2020 without any composition shifts. To achieve this effect, I multiplied Biden’s margin rate within each group by a factor of .9, preserving the relative rates at which each group voted for Biden, but reducing his vote totals. As can be seen, this shift is sufficient to shift just over 17.2 million votes to Trump versus the actual reported result.

Conversely, Table A5 demonstrates how composition would have had to shift, without any changes in the underlying rates within groups, in order to deliver a Trump

Table A4: Hypothetical Data, Only Rate Changes

Election	Group	Cand. 1	Cand. 2	Total	Gross Mar.	Mar. Rate (%)	Comp. (%)	
1st Election (2020)	Black	15.42	1.47	16.89	+13.95	+82.60	10.89	Votes
	Hispanic	13.21	4.47	17.68	+8.73	+49.40	11.41	
	Other	8.64	5.16	13.80	+3.48	+25.20	8.90	
	White	47.95	57.21	105.16	-9.25	-8.80	67.84	
	NA	0.80	0.67	1.47	+0.13	+8.60	0.95	
	Total	86.02	68.98	155.00	+17.03	+10.99	100.00	
2nd Election (2020)	Black	13.88	3.01	16.89	+10.86	+64.34	10.89	
	Hispanic	11.89	5.79	17.68	+6.09	+34.46	11.41	
	Other	7.78	6.03	13.80	+1.75	+12.68	8.90	
	White	43.16	62.00	105.16	-18.84	-17.92	67.84	
	NA	0.72	0.75	1.47	-0.03	-2.26	0.95	
	Total	77.41	77.59	155.00	-0.17	-0.11	100.00	
Comparing the Two Elections	Black	-1.54	1.54		-3.08	-18.26	0.00	
	Hispanic	-1.32	1.32		-2.64	-14.94	0.00	
	Other	-0.86	0.86		-1.73	-12.52	0.00	
	White	-4.80	4.80		-9.59	-9.12	0.00	
	NA	-0.08	0.08		-0.16	-10.86	0.00	
	Total	-8.60	8.60		-17.20	-11.10	0.00	

are reported in millions.

popular vote win. In this specification, to achieve the desired result, I had to multiply the size of each of the non-White voting blocs by a factor of .44, essentially reducing them by more than half. This leads to a shift in the composition of the electorate from just under 68% White to nearly 86% White, an enormous shift. The impossibility of this shift highlights the importance that Republicans will need to place on changing rates within minority groups in future elections, especially as demographic changes continue to accelerate in the future. However, as shown in Section 4, there is significant reason to believe that Republicans have succeeded at changing rates in recent elections, suggesting that demographics driving elections towards Democrats is not an assured outcome.

I have not produced a table comparing the RCVD approach and the LDD for either Table A4 or Table A5, as either specification produces identical results.

Table A5: Hypothetical Data, Only Composition Changes

Election	Group	Cand. 1	Cand. 2	Total	Gross Mar.	Mar. Rate (%)	Comp. (%)	
1st Election (2020)	Black	15.42	1.47	16.89	+13.95	+82.60	10.89	Votes
	Hispanic	13.21	4.47	17.68	+8.73	+49.40	11.41	
	Other	8.64	5.16	13.80	+3.48	+25.20	8.90	
	White	47.95	57.21	105.16	-9.25	-8.80	67.84	
	NA	0.80	0.67	1.47	+0.13	+8.60	0.95	
	Total	86.02	68.98	155.00	+17.03	+10.99	100.00	
2nd Election (2020)	Black	6.78	0.65	7.43	+6.14	+82.60	4.79	Votes
	Hispanic	5.81	1.97	7.78	+3.84	+49.40	5.02	
	Other	3.80	2.27	6.07	+1.53	+25.20	3.92	
	White	60.68	72.39	133.07	-11.71	-8.80	85.85	
	NA	0.35	0.30	0.65	+0.06	+8.60	0.42	
	Total	77.43	77.57	155.00	-0.14	-0.09	100.00	
Comparing the Two Elec- tions	Black	-8.63	-0.82	-9.46	-7.81	0.00	-6.10	
	Hispanic	-7.40	-2.50	-9.90	-4.89	0.00	-6.39	
	Other	-4.84	-2.89	-7.73	-1.95	0.00	-4.99	
	White	12.73	15.18	27.91	-2.46	0.00	+18.01	
	NA	-0.45	-0.38	-0.83	-0.07	0.00	-0.53	
	Total	-8.59	8.59	0.00	-17.18	-11.08	0.00	

are reported in millions.

Rate and Composition Shift Data

Robustness

To demonstrate the robustness of the method, I present Appendix Tables A7 and A8. Table A7 splits votes out from the Non-White category into Black and All Others, leaving the White category untouched. Table A8 shows that both the LDD as well as the RCVD approach correctly leave the specification of the impact of rate, composition, and volume to White voters (the unchanged category) unchanged from Table A8. This highlights the robustness to irrelevant alternatives of the RCVD approach, showing how it behaves comparably to the LDD.

Table A6: Hypothetical Data, Both Rate and Composition

Method	Group	Rate	Comp.	Volume	Calc Mar Chng	Var to Act	Var. %	
Derivative Calculation	Black	-2.16	-2.79	0	-4.95	-0.43	9.6	Votes
	Hispanic	-1.85	-1.75	0	-3.60	-0.37	11.5	
	Other	-1.21	-0.70	0	-1.91	-0.24	14.5	
	White	-6.71	-0.88	0	-7.59	+0.64	-7.7	
	NA	-0.11	-0.03	0	-0.14	-0.02	19.5	
	Total	-12.04	-6.13	0	-18.18	-0.43	2.4	
RCV Calculation	Black	-2.16	-2.36	0	-4.52	0.00	0.0	
	Hispanic	-1.85	-1.38	0	-3.23	0.00	0.0	
	Other	-1.21	-0.45	0	-1.66	0.00	0.0	
	White	-6.71	-1.51	0	-8.23	0.00	0.0	
	NA	-0.11	-0.00	0	-0.11	0.00	0.0	
	Total	-12.04	-5.71	0	-17.75	0.00	0.0	
Comparing the Two Ap- proaches	Black		-0.43	0	-0.43			
	Hispanic		-0.37	0	-0.37			
	Other		-0.24	0	-0.24			
	White		+0.64	0	+0.64			
	NA		-0.02	0	-0.02			
	Total		-0.43	0	-0.43			

are reported in millions. Variance percentages are shown with one decimal place.

Table A7: Comparing Two Elections, Robustness to Irrelevant Alternatives

Election	Group	Cand. 1	Cand. 2	Total	Margin	Mar. Rate (%)	Comp. (%)
1st Election (2016)	Black	13.25	0.90	14.15	+12.35	+87.3	11.0
	All Others	16.25	6.60	22.85	+9.65	+42.2	17.7
	White	39.00	53.00	92.00	-14.00	-15.2	71.3
	Total	68.50	60.50	129.00	+8.00	+6.2	100.0
2nd Election (2020)	Black	15.40	1.50	16.90	+13.90	+82.2	10.9
	All Others	22.60	10.50	33.10	+12.10	+36.6	21.4
	White	48.00	57.00	105.00	-9.00	-8.6	67.7
	Total	86.00	69.00	155.00	+17.00	+11.0	100.0
Comparing the Two Elec- tions	Black	2.15	0.60	2.75	+1.55	-5.0	-0.1
	All Others	6.35	3.90	10.25	+2.45	-5.7	+3.6
	White	9.00	4.00	13.00	+5.00	+6.6	-3.6
	Total	17.50	8.50	26.00	+9.00	+4.8	0.0

Votes are reported in millions. Candidate 1 refers to Clinton (2016) and Biden (2020); Candidate 2 refers to Trump.

Table A8: Concept Demonstration, Robustness to Irrelevant Alternatives

Method	Group	Rate	Comp.	Volume	Calc Mar Chng	Var to Act	Var. %	
Derivative Calculation	Black	−0.71	−0.07	2.49	1.70	0.15	+9.9	Votes
	All Others	−1.30	1.98	1.94	2.63	0.18	+7.4	
	White	6.11	0.70	−2.82	3.99	−1.01	−20.1	
	Total	4.11	2.61	1.61	8.33	−0.67	−7.4	
RCV Calculation	Black	−0.71	−0.07	2.33	1.55	0.00	0.0	
	All Others	−1.30	1.72	2.03	2.45	0.00	0.0	
	White	6.11	0.40	−1.51	5.00	0.00	0.0	
	Total	4.11	2.04	2.85	9.00	0.00	0.0	
Comparing the Two Ap- proaches	Black		−0.00	0.16	0.15			
	All Others		0.27	−0.08	0.18			
	White		0.31	−1.31	−1.01			
	Total		0.57	−1.24	−0.67			

are reported in millions. Variance percentages are shown with one decimal place.

Table A9: Concept Demonstration for Trump, Real Data from Marble et al. (2024)

Method	Group	Rate	Comp.	Volume	Calc Mar Chng	Var to Act	Var. %	
Derivative Calculation	Black	0.68	0.04	−0.82	−0.11	1.48	−93.1	Votes are
	Hispanic	0.26	−0.40	−0.45	−0.59	1.44	−71.0	
	Other	0.41	−0.35	−0.18	−0.11	0.69	−85.8	
	White	−5.52	−0.31	0.91	−4.92	−0.56	+12.9	
	NA	0.02	−0.02	−0.01	−0.00	0.03	−86.4	
	Total	−4.14	−1.04	−0.55	−5.73	3.07	−34.9	
RCV Calculation	Black	0.68	0.07	−2.34	−1.59	0.00	0.0	
	Hispanic	0.26	−0.83	−1.46	−2.03	0.00	0.0	
	Other	0.41	−0.63	−0.58	−0.80	0.00	0.0	
	White	−5.52	−0.39	1.55	−4.35	0.00	0.0	
	NA	0.02	−0.03	−0.02	−0.03	0.00	0.0	
	Total	−4.14	−1.81	−2.86	−8.81	0.00	0.0	
Comparing the Two Ap- proaches	Black		−0.04	1.52	1.48			
	Hispanic		0.43	1.02	1.44			
	Other		0.28	0.40	0.69			
	White		0.08	−0.65	−0.56			
	NA		0.01	0.01	0.03			
	Total		0.77	2.31	3.07			

reported in millions. Variance percentages are shown with one decimal place.